Sinais e Sistemas para Engenharia

MÓDULO 10 - SÉRIE DE FOURIER – TF DE SINAIS PERIÓDICOS

VERSÃO OCTAVE

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Podemos gerar um Sinal Periódico a partir da repetição de um sinal Não-Periódico

```
function y=MakePeriodic(f_np,t,t_ini,t_fim)
    tp = mod((t-t_ini),(t_fim-t_ini))+t_ini;
    index = find(tp<t_ini);
    if !isempty(index)
      tp(index)=tp(index) + (t_fim-t_ini);
    end
    y = f_np(tp);
endfunction
```

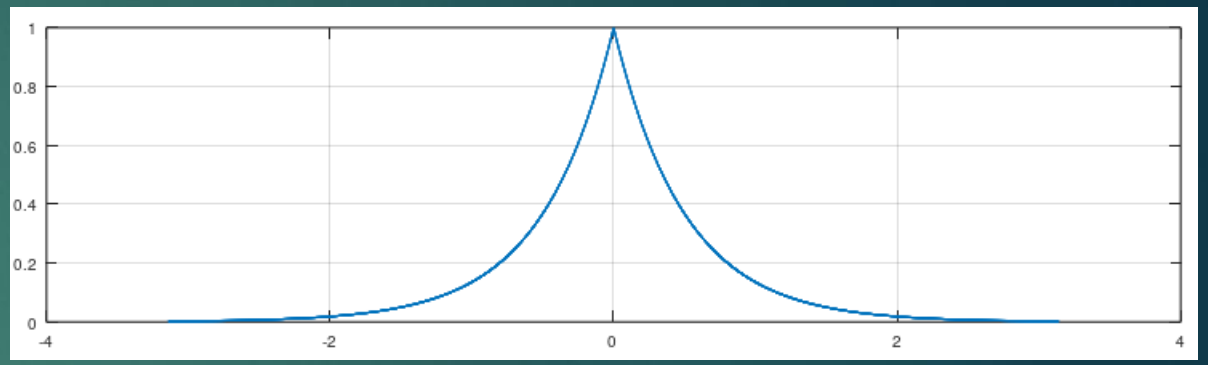
Exemplo 1) Como exemplo, vamos analisar o sinal periódico de período $T = 2\pi$, sendo cada período constituído de exponencial decrescente $y(t) = e^{-2|t|}, -\pi < t < \pi$

$$w_o = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

1 período

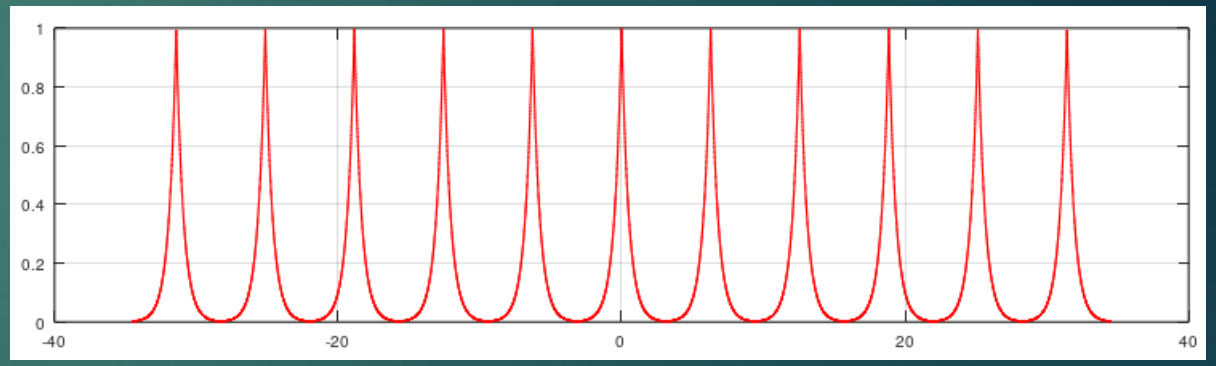
```
>> f1 = @(t) exp(-2*abs(t));
>> t=linspace(-pi,pi,10001);
>> plot(t,f1(t));
```

```
>> t=linspace(-pi,pi,10001);
>> plot(t,MakePeriodic(f1,t,-pi,pi),'r');
```



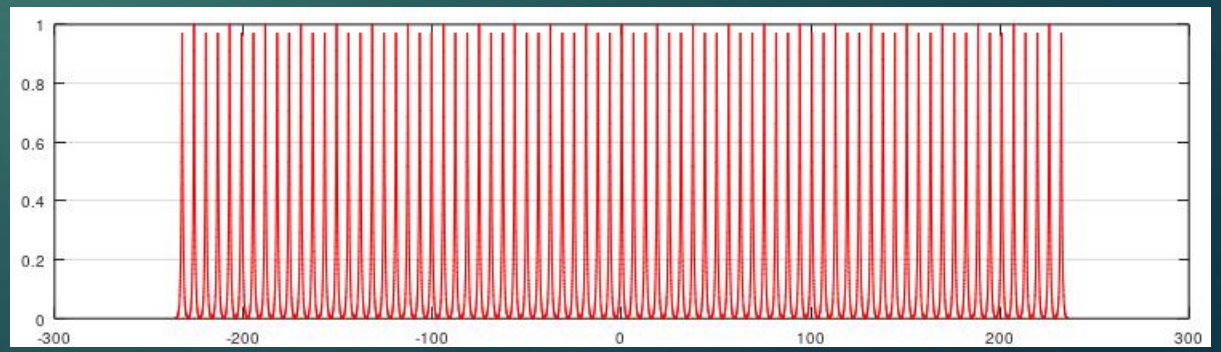
11 períodos

```
>> t=linspace(-11*pi,11*pi,10001);
>> plot(t,MakePeriodic(f1,t,-pi,pi),'r');
```



75 períodos

```
>> t=linspace(-75*pi,75*pi,10001);
>> plot(t,MakePeriodic(f1,t,-pi,pi),'r');
```



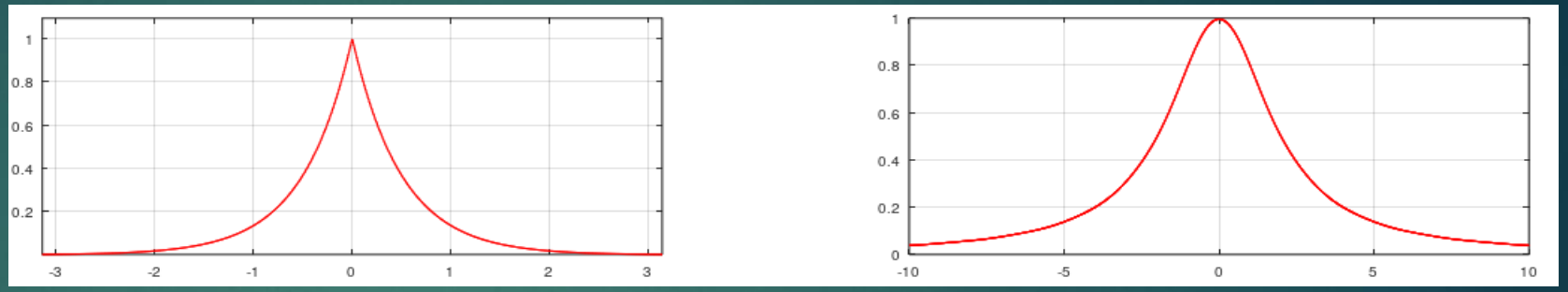
Transformada de Fourier de N Períodos deste Sinal

$$y(t) = e^{-2|t|}, -\pi < t < \pi$$

```
>> f1 = @(t) exp(-2*abs(t));
>> f_T = @ (t) MakePeriodic(f1,t,-pi,pi);
```

```
>> PlotTF(f_T,10,-pi,pi,'r')
```

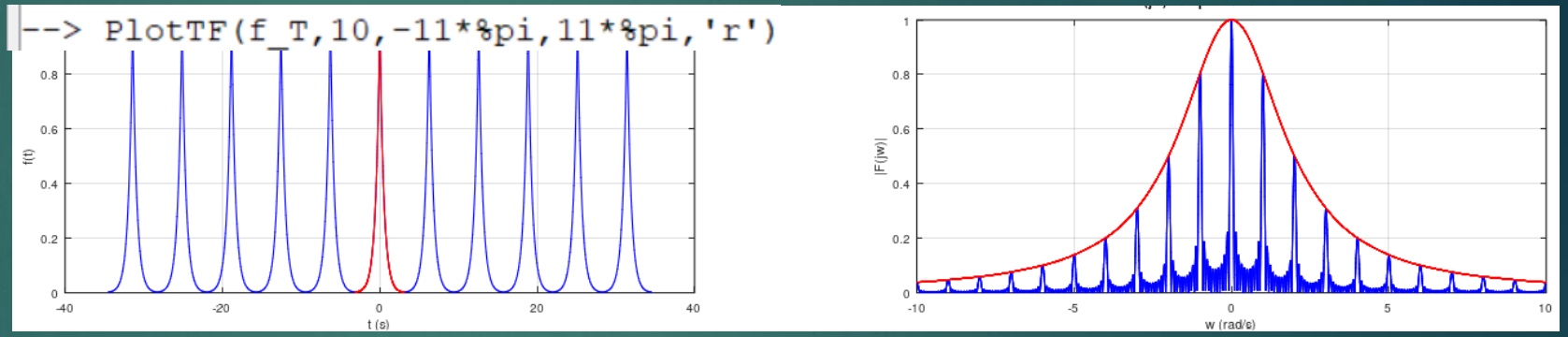
1 período da Exponencial
$y(t) = e^{-2|t|}, -\pi < t < \pi$ |



```
>> PlotTF(f_T,10,-11*pi,11*pi,'b')
```

11 períodos
$T = 2\pi$

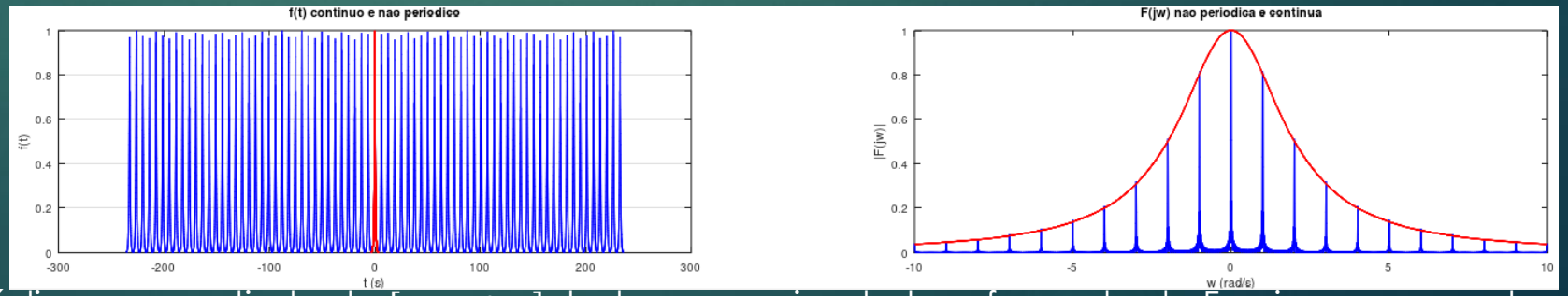
$$w_o = \frac{2\pi}{T} = 1$$




```
>> PlotTF(f_T,10,-75*pi,75*pi,'b')
```

75 períodos
$T = 2\pi$

$$w_o = \frac{2\pi}{T} = 1$$



No limite, quando o sinal se torna periódico, se repetindo de $[-\infty, +\infty]$, toda a energia da transformada de Fourier se concentra nas frequências múltiplas da frequência fundamental do sinal periódico $w_o = \left(\frac{2\pi}{T}\right) = 1$, de modo que a transformada se torna discreta.

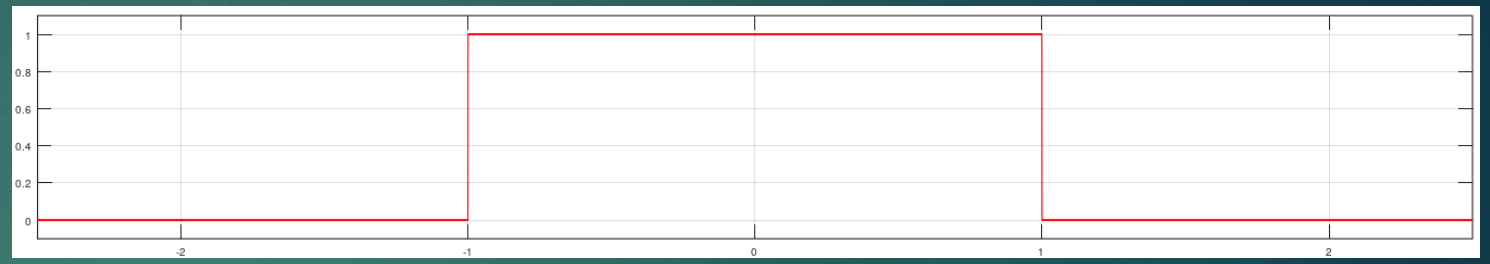
Exemplo 2) analisar o sinal periódico de período $T = 5$, sendo cada período constituído de um pulso unitário $y(t) = 1, -1 < t < 1$

```
--> deff('y=f2(t)','y=f_u(t+1)-f_u(t-1)');
```

1 período

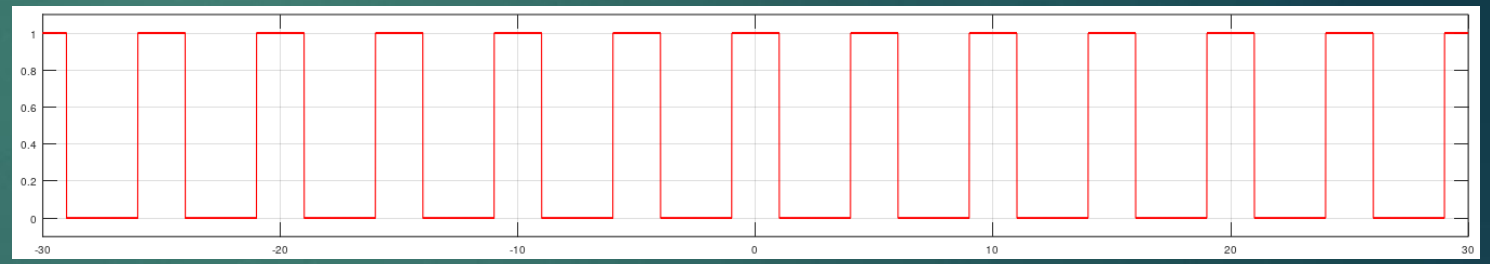
$$w_o = \frac{2\pi}{T} = \frac{2\pi}{5} = 1.26$$

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> t=linspace(-2.5,2.5,10001);
>> plot(t,MakePeriodic(f2,t,-2.5,2.5),'r');
```



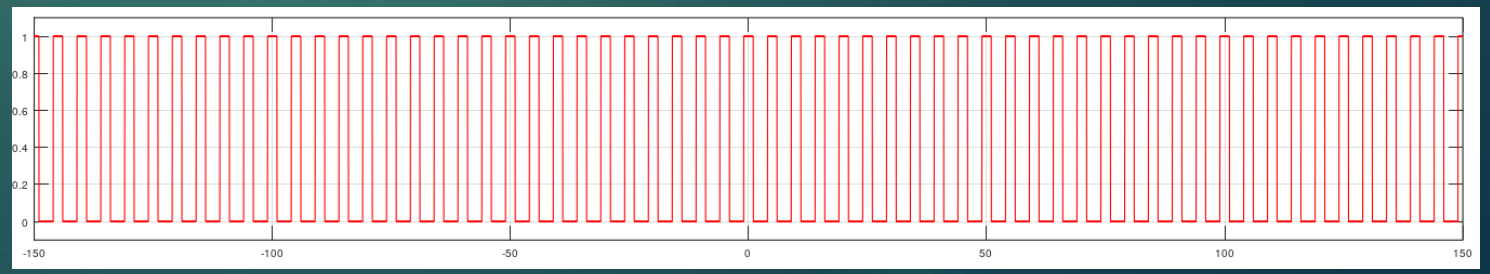
12 períodos

```
>> t=linspace(-30,30,10001);
>> plot(t,MakePeriodic(f2,t,-2.5,2.5),'r');
```



60 períodos

```
>> t=linspace(-150,150,10001);
>> plot(t,MakePeriodic(f2,t,-2.5,2.5),'r');
```



Transformada de Fourier de N Períodos de um Pulso unitário, largura 1, T=5

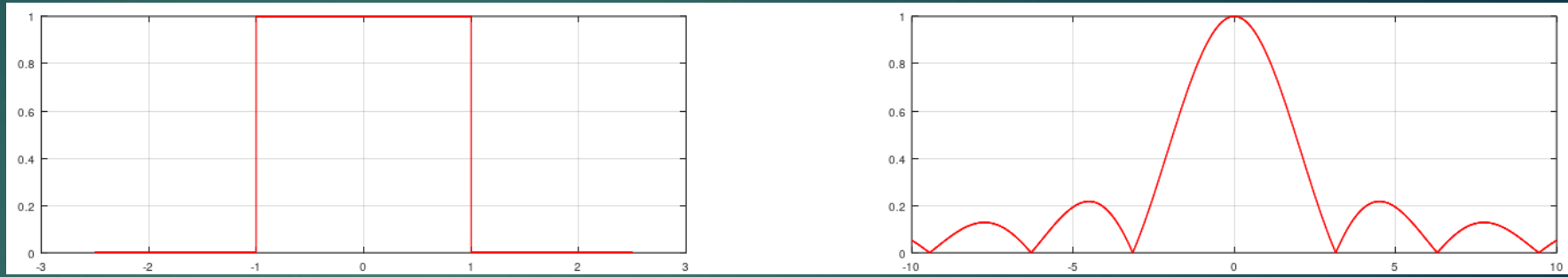
$$y(t) = \begin{cases} 1; & -1 \le t \le 1 \\ 0; & caso\ contráio \end{cases}; \quad T = 5$$

$$w_o = \frac{2\pi}{T} = 1.26$$

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> f_T = @ (t) MakePeriodic(f2,t,-2.5,2.5);
```

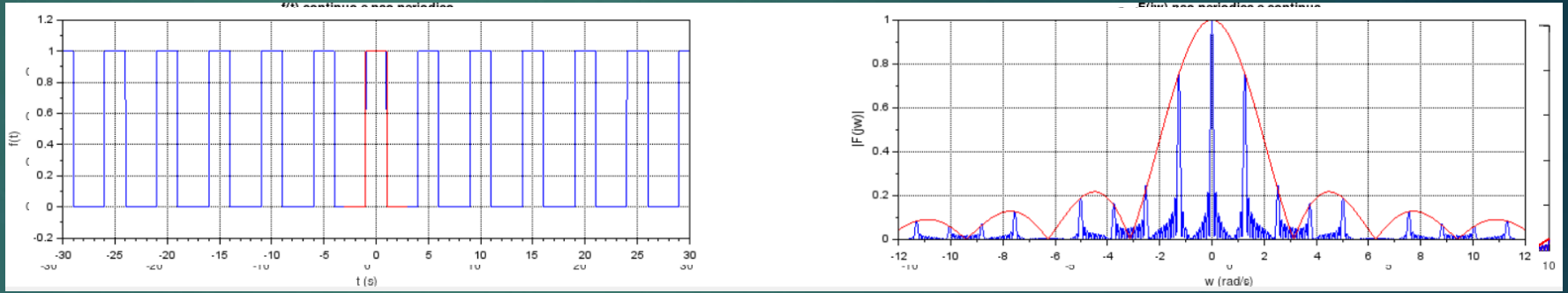
1 período

```
>> PlotTF(f_T,10,-2.5,2.5,'r')
```



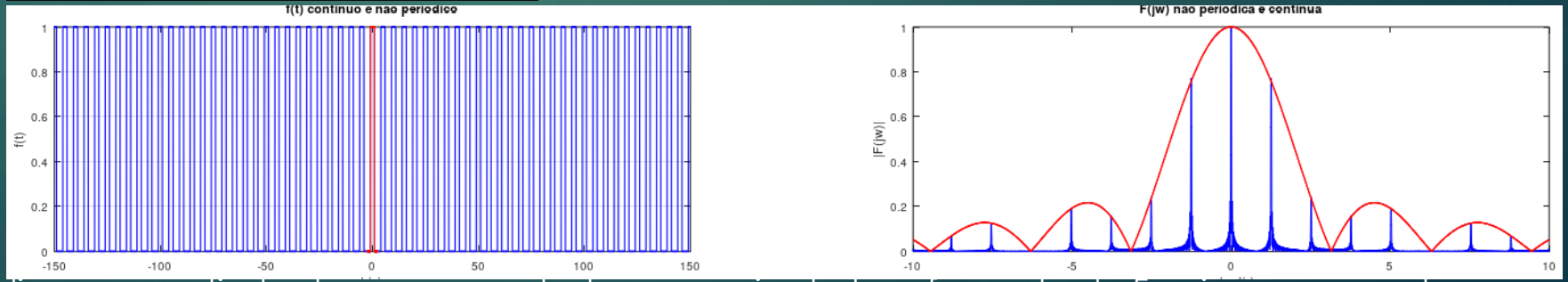
12 períodos

```
>> PlotTF(f_T,10,-30,30,'b')
```



60 períodos

```
>> PlotTF(f_T,10,-150,150,'b')
```



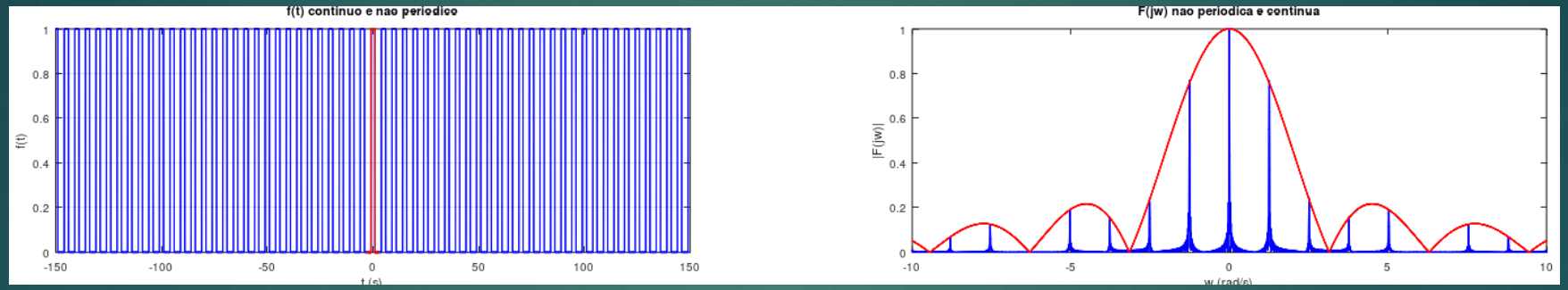
No limite, quando o sinal se torna periódico, se repetindo de $-\infty$ a $+\infty$, toda a energia da transformada de Fourier se concentra nas frequências múltiplas da frequência fundamental do sinal periódico $w_o = \left(\frac{2\pi}{T}\right) = 1.26$

Transformada de Fourier de N Períodos de um Sinal

Vamos comparar o sinal pulso unitário periódico de período T=5; com o sinal pulso unitário periódico de período 10

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> f_T = @ (t) MakePeriodic(f2,t,-2.5,2.5);
>> PlotTF(f_T,10,-150,150,'b')
```

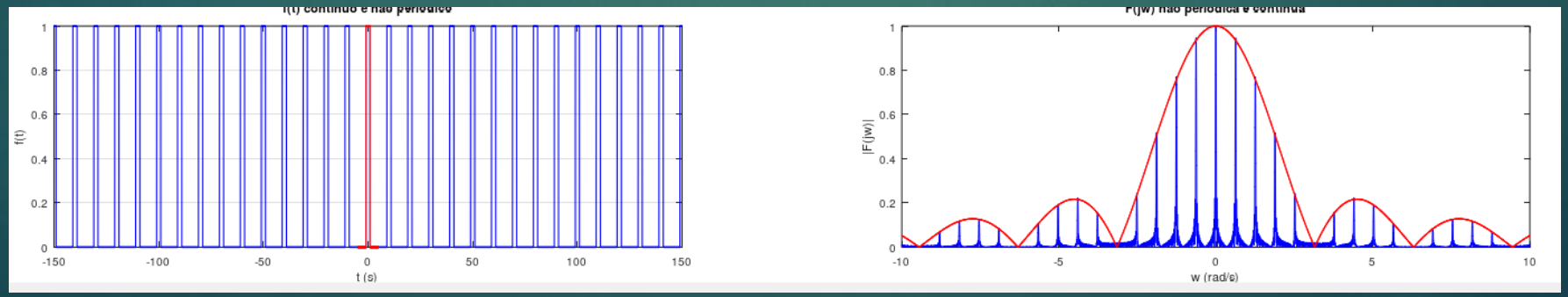
60 períodos
$T = 5$



$$w_o = \frac{2\pi}{T} = 1.26$$

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> f_T = @ (t) MakePeriodic(f2,t,-5,5);
>> PlotTF(f_T,10,-150,150,'b')
```

30 períodos
$T = 10$

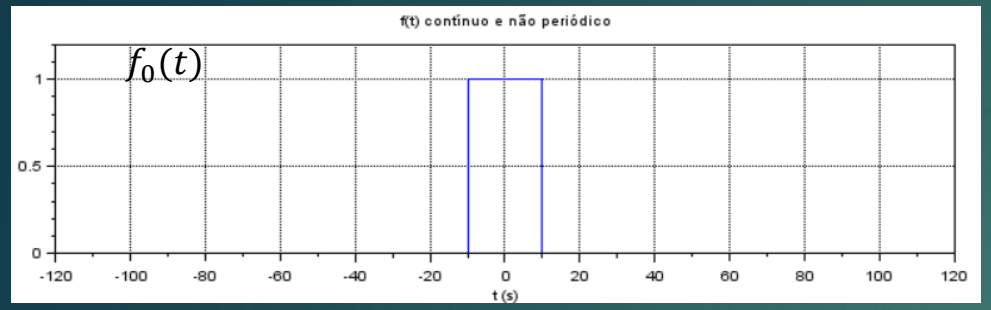


$$w_o = \frac{2\pi}{T} = 0.63$$

Notar que quando dobramos os período no tempo, reduzimos pela metade o espaçamento das amostras na frequência.

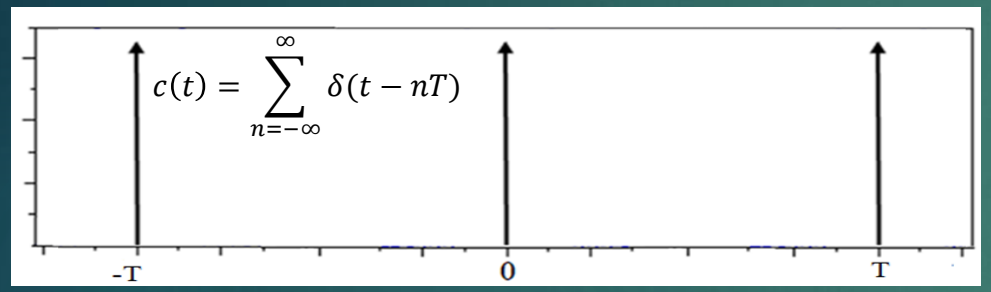
Matematicamente, podemos criar um sinal periódico $f_T(t)$, com período T, a partir de um sinal não periódico $f_0(t)$. Para isso basta convoluirmos $f_0(t)$ com um trem de impulso $c(t)$ com período T

Tempo



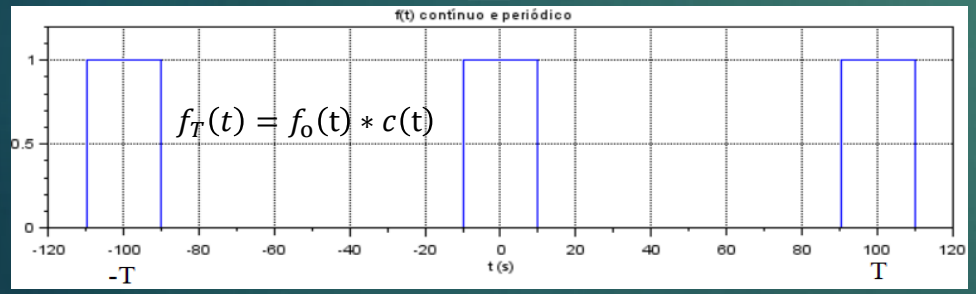
TRANSFORMADA DE FOURIER

$*$



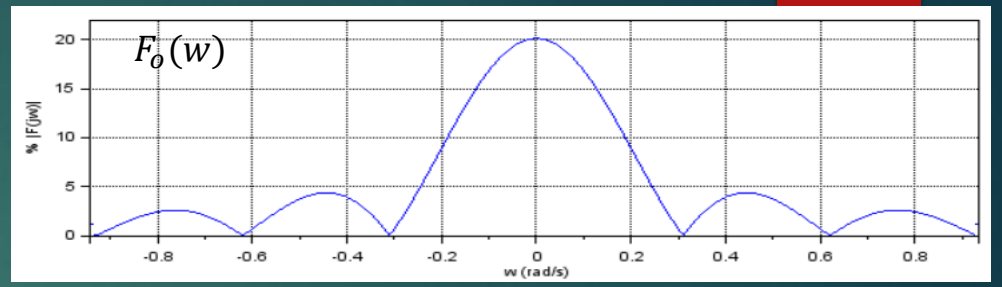
TRANSFORMADA DE FOURIER

$=$

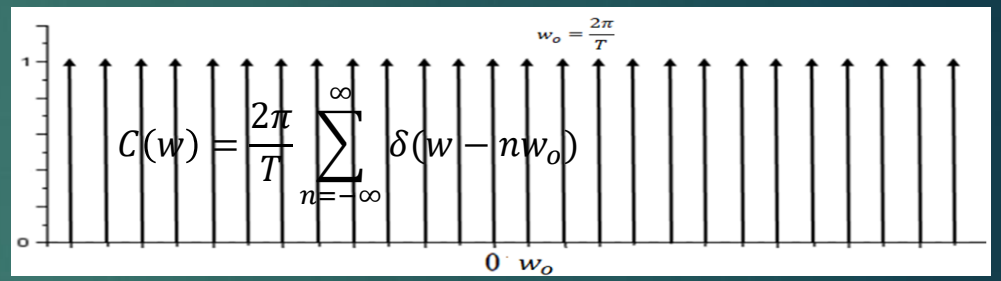


TRANSFORMADA DE FOURIER

Frequência



X



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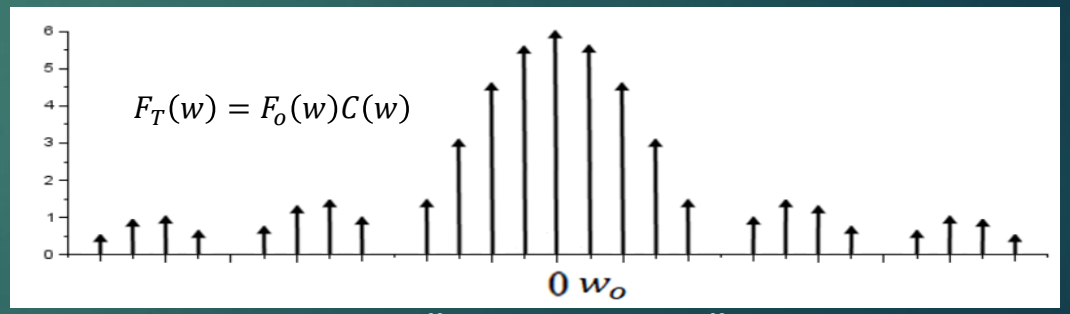


$$F_T(w) = F_o(w)C(w) = F_o(w)\frac{2\pi}{T}\sum_{n=-\infty}^{\infty}\delta(w - nw_o) = \sum_{n=-\infty}^{\infty}\frac{2\pi}{T}F_o(nw_o)\delta(w - nw_o)$$

A convolução no tempo por um trem de impulsos espaçados de T, equivale reproduzir o sinal original a cada T segundos, isto é, o sinal se torna periódico.

Notar que a transformada de um trem de impulsos espaçados de t_o é também um trem de impulsos (na frequência), porém espaçados de w $\frac{2\pi}{t_o}$, com amplitude variável $\frac{2\pi}{T}F_o(nw_o)$

Série de Fourier
Um caso especial da Transformada de Fourier

- A **Série de Fourier** pode ser interpretada com sendo um caso especial da **Transformada de Fourier** quando o sinal no tempo $f_T(t)$ é contínuo e periódico com período T (o subíndice T indica periodicidade em T) . Neste caso a Transformada $\mathrm{F_T}(jw)$ deste sinal será discreta e não periódica. Por ser discreta, trocamos a frequência contínua $'jw'$ pelo índice discreto $'n'$, por isso a série de Fourier será representada pelo sinal discreto $F[n]$
- Com a Série de Fourier , podemos decompor o sinal original periódico $f_T(t)$, com período T (isto é, com frequência fundamental $w_o = \frac{2\pi}{T}$) , em um soma infinita de exponenciais complexas com frequência múltiplas inteiras da frequência w_o (que chamamos de harmônicos)

$$\boldsymbol{f_T(t)} \xRightarrow{\textbf{Série de Fourier (Análise)}} \boldsymbol{F}[n]$$

Separar o sinal periódico $\boldsymbol{f_T(t)}$ nos seus harmônicos F(n), espaçados de $\mathrm{w_o} = \frac{2\pi}{T}$

$$\boxed{F[n] = \frac{1}{T}\int_0^T f(t)e^{-jnw_o t}dt}$$

$$w_o = \frac{2\pi}{T}$$

$$\boldsymbol{F}[n] \xRightarrow{(\textbf{Série de Fourier})^{-1}=\textbf{Síntese}} \boldsymbol{f_T(t)}$$

Reconstituir o sinal periódico $\boldsymbol{f_T(t)}$ a partir dos harmônicos F(n) - soma infinita de exponenciais complexas com frequência múltiplas inteiras da frequência $w_o = \frac{2\pi}{T}$.

$$\boxed{f(t) = \sum_{n=-\infty}^{\infty} F[n]e^{jnw_o t}}$$

Dedução do cálculo analítico da Série de Fourier

Vimos que a Transformada de Fourier de um sinal Periódico $f_T(t)$ com período $T = \frac{2\pi}{w_o}$ é um trem de impulsos $C(w)$ espaçados de w_o no domínio da frequência, sendo que cada impulso tem energia $\frac{2\pi}{T}F_o(nw_o)$, onde $F_o(w)$ é a transformada de Fourier $f_0(t)$ (que nada mais é que um período de $f_T(t)$) .

$$f_T(t) = f_o(\mathrm{t}) * c(t) \overset{F}{\Longrightarrow} F_T(w) = F_o(w)C(w)$$

$$f_T(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F_T(w)e^{jwt}dw = \frac{1}{2\pi}\int_{-\infty}^{\infty} F_o(w)C(w)e^{jwt}dw \quad ; mas, C(w) = \frac{2\pi}{T}\sum_{n=-\infty}^{\infty}\delta(w - nw_o)$$

$$f_T(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F_o(w)\left[\frac{2\pi}{T}\sum_{n=-\infty}^{\infty}\delta(w - nw_o)\right]e^{jwt}dw = \sum_{n=-\infty}^{\infty}\frac{1}{T}F_o(nw_o)e^{nw_ot}\int_{-\infty}^{\infty}\delta(w - nw_o)dw = \sum_{n=-\infty}^{\infty}\frac{1}{T}F_o(nw_o)e^{nw_ot}$$

$Se\ \ definirmos\ :\quad F[n] = \frac{1}{T}F_o(nw_o);$ então:

$$F[n] = \frac{1}{T}F_o(nw_o) = \frac{1}{T}\int_{-\infty}^{\infty} f_0(t)e^{-jnw_o}dt \qquad \boldsymbol{F[n] = \frac{1}{T}\int_0^T f_T(t)e^{-jnw_o}dt} \quad (\boldsymbol{I})$$

À sequencia discreta $F(n)$ damos o nome de Série de Fourier. $F(n)$ nada mais é do que o valor das amplitudes dos de exponenciais complexas com frequência nw_o (múltiplas inteiras da frequência w_o).

Deste modo , sinal original $f_T(w)$ (Equação I) pode ser reconstituído a partir da sequência discreta F(n) , pela soma ponderada destas exponenciais complexas, com frequência múltiplas inteiras da frequência w_o.

$$f_T(t) = \sum_{n=-\infty}^{\infty}\frac{1}{T}F_o(nw_o)e^{nw_ot} \qquad f_T(t) = \sum_{n=-\infty}^{\infty}F[n]e^{nw_ot} \quad (II)$$

Exemplo 1 – Trem de Pulsos

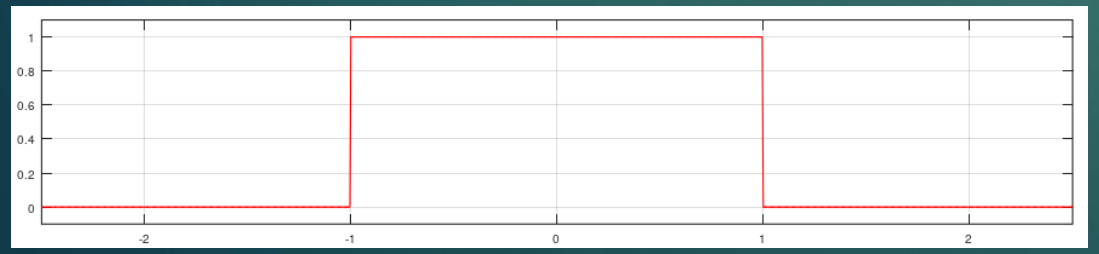
$$y(t) = \begin{cases} 1; & -1 \leq t \leq 1 \\ 0; & caso\ contráio \end{cases}; \quad T = 5$$

Notar que f(t) é periódica com período 5

$w_o = \frac{2\pi}{5}$=1.257

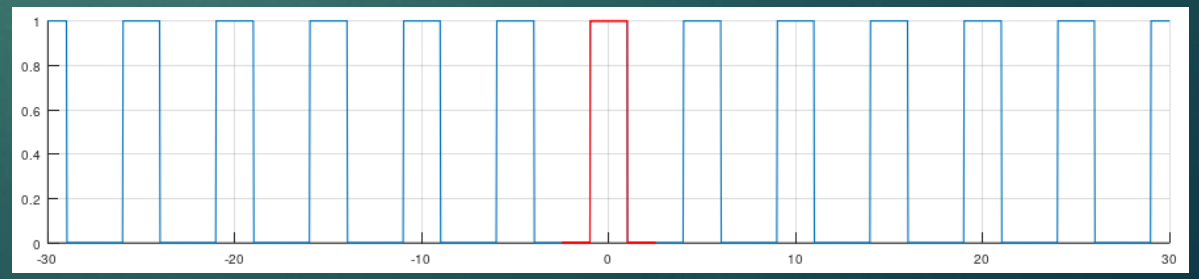
Definir f(t) em 1 período e plotar 1 período

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> t=linspace(-2.5,2.5,1001);
>> plot(t,f2(t),'r');
```



Replicar a função em em um eixo t qualquer, com MakePeriodic()

```
>> f_T = @ (t) MakePeriodic(f2,t,-2.5,2.5);
>> tp=linspace(-30,30,2000);
>> hold on; grid on;
>> plot(tp,f_T(tp));
>> plot(t,f2(t),'r');
```



Cálculo Serie com Coeficiente Complexos, através de 1 integral complexa

$$y(t) = \begin{cases} 1; & -1 \le t \le 1 \\ 0; & caso\ contráio \end{cases}; \quad T = 5$$

Notar que f(t) é periódica com período 5

$$w_o = \frac{2\pi}{5}$$

Cálculo analítico da Série:

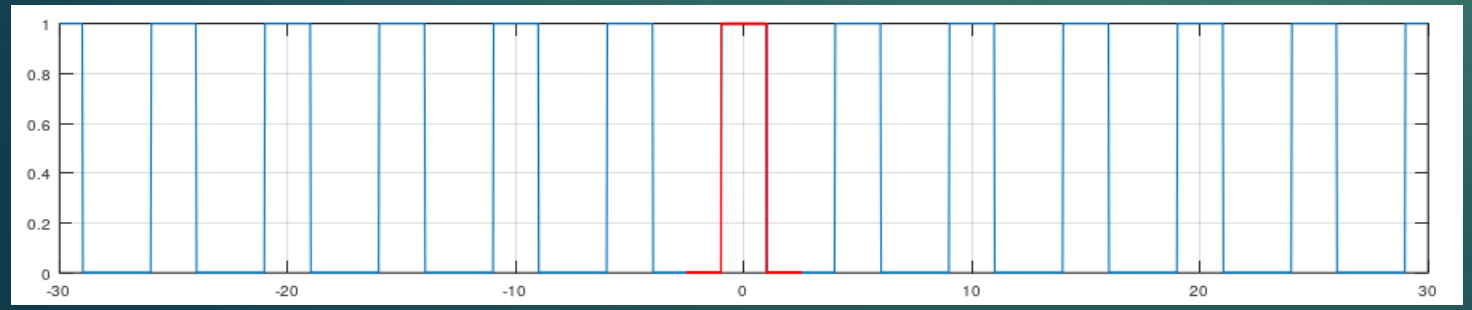
$$F[n] = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)e^{(-jnw_o t)}dt = \frac{1}{5}\left[\int_{-1}^{1}(1)e^{\left(-jn\frac{2\pi}{5}t\right)}dt\right] = \frac{\sin\left(\frac{2\pi n}{5}\right)}{n\,\pi} = \frac{2}{5}\mathrm{sinc}\left(\frac{2n}{5}\right)$$

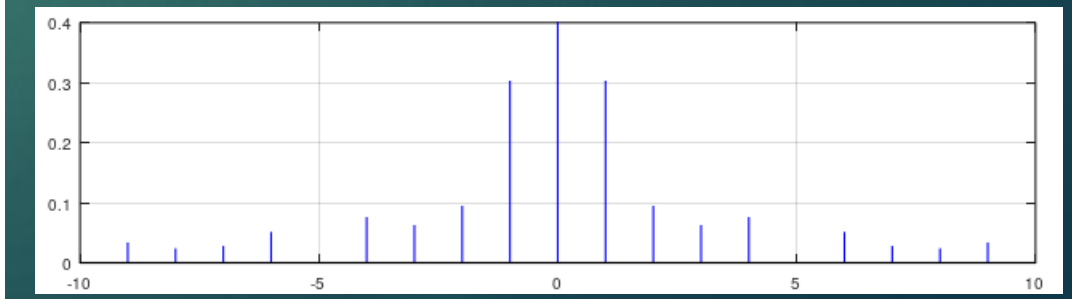
```
>> F = @(n) 2/5*sinc(2*n/5);
>> n=[-9:9];
>> [n' F(n)']
ans =

  -9.0000  -0.0336
  -8.0000  -0.0234
  -7.0000   0.0267
  -6.0000   0.0505
  -5.0000  -0.0000
  -4.0000  -0.0757
  -3.0000  -0.0624
  -2.0000   0.0935
  -1.0000   0.3027
        0   0.4000
   1.0000   0.3027
   2.0000   0.0935
   3.0000  -0.0624
   4.0000  -0.0757
   5.0000  -0.0000
   6.0000   0.0505
   7.0000   0.0267
   8.0000  -0.0234
   9.0000  -0.0336
```

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> f_T = @ (t) MakePeriodic(f2,t,-2.5,2.5);
>> t=linspace(-2.5,2.5,1001);
>> tp=linspace(-30,30,2000);
>> hold on; grid on;
>> plot(tp,f_T(tp));
>> plot(t,f2(t),'r');
```

```
>> F2 = @(t) 2/5*sinc(2*t/5);
>> n=[-9:9];
>> stem(n,abs(F2(n)),'b','marker','none');
```





Os coeficiente complexos F(n) numericamente a partir de 2 integrais reais

$$F[n] = \frac{1}{T}\int_0^T f(t)e^{-jnw_o}dt = \frac{1}{T}\int_0^T f(t)\cos(nw_o)dt - j\frac{1}{T}\int_0^T f(t)\sin(nw_o)\,dt$$

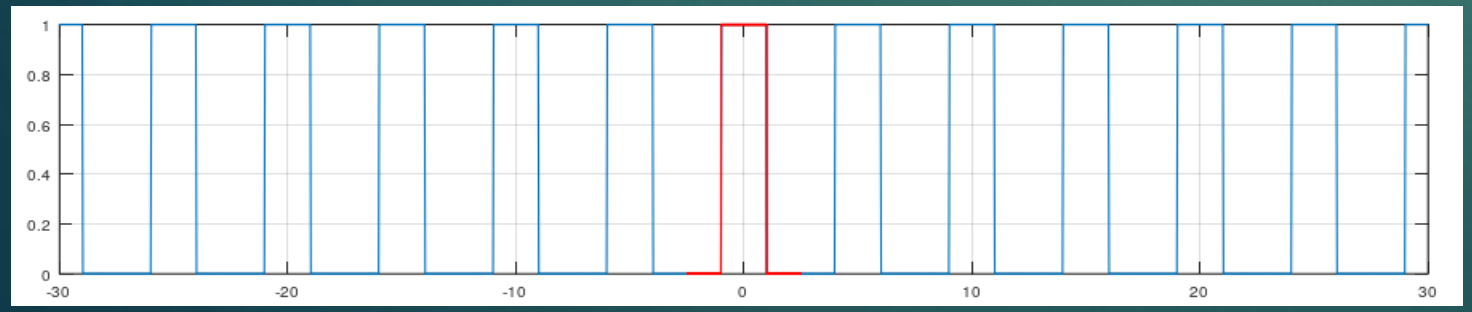
$$\boldsymbol{Se \quad a_n = \frac{2}{T}\int_0^T f(t)\cos(nw_o t)\,dt \quad ; \quad n \geq 0}$$

$$\boldsymbol{e \quad b_n = \frac{2}{T}\int_0^T f(t)\,\mathrm{sen}(nw_o t)\,dt; \qquad n \geq 0}$$

$$Então: \quad F[n] = \begin{cases} \dfrac{a_n - jb_n}{2} & n \geq 0 \\ \dfrac{a_n + jb_n}{2}; & n < 0 \end{cases}$$

```
function [F,n]=SerieFourier(f_T,t_ini,t_fim,N)
    T=(t_fim-t_ini);
    w0 = 2* pi  /T;
    C=[];
    for n=1:N
        fsin= @(t) f_T(t).*sin((n-1)*w0*t);
        fcos= @(t) f_T(t).*cos((n-1)*w0*t);
        A=2/T*intg_simpson(fcos,t_ini,t_fim,n);
        B=2/T*intg_simpson(fsin,t_ini,t_fim,n);
        C=[C,(A- i *B)/2];
    end
  %  C=clean(C,1e-10);%zera valores pequenos
    F=[flip(conj(C)) C(2:N)];
    n=[-N+1:N-1];
endfunction
```

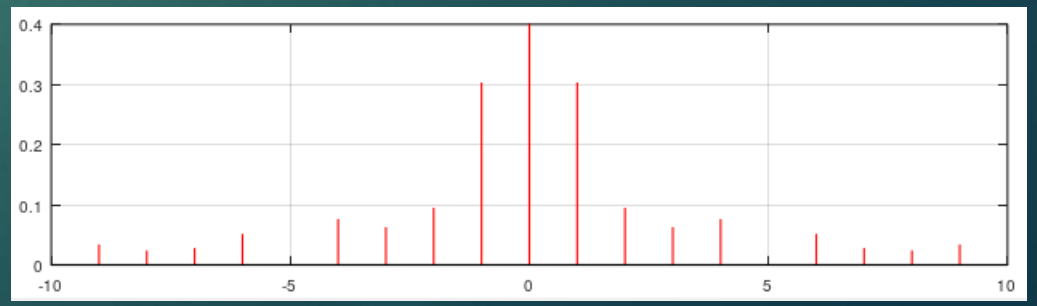
```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> f_T = @ (t) MakePeriodic(f2,t,-2.5,2.5);
>> t=linspace(-2.5,2.5,1001);
>> tp=linspace(-30,30,2000);
>> hold on; grid on;
>> plot(tp,f_T(tp));
>> plot(t,f2(t),'r');
```



Reconstituição do sinal original:

$$f(t) = \sum_{n=-\infty}^{\infty} F[n]e^{nw_o t} = \frac{a_o}{2} + \sum_{1}^{\infty}\left(a_n\cos(nw_o t) + jb_n sen(nw_o t)\right)$$

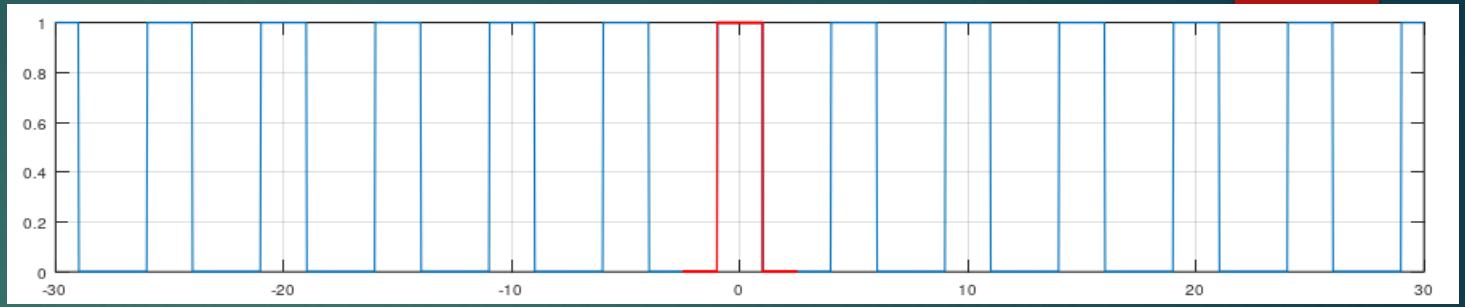
```
>> Fn=SerieFourier(f2,-2.5,2.5,10);
>> stem(n,abs(Fn),'r','marker','none');
```



Síntese – Reconstrução de $f_T(t)$ através dos coeficientes Complexos

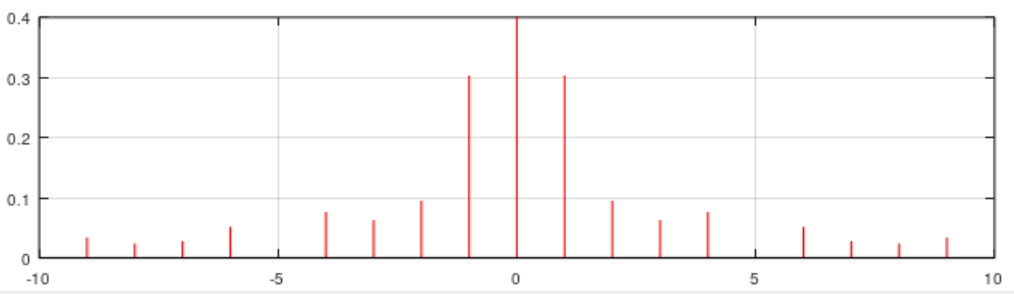
Começamos com a função periódica $f(t)$

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> f_T = @ (t) MakePeriodic(f2,t,-2.5,2.5);
>> t=linspace(-2.5,2.5,1001);
>> tp=linspace(-30,30,2000);
>> hold on; grid on;
>> plot(tp,f_T(tp));
>> plot(t,f2(t),'r');
```



Calculamos n coeficiente da séria F(n)

$$f(t) = \begin{cases} 1; & -1 \le t \le 1 \\ 0; & caso\ contráio \end{cases};\ T = 5 \qquad F[n] = \frac{1}{T}\int_0^T f(t)e^{-jnw_o}dt$$



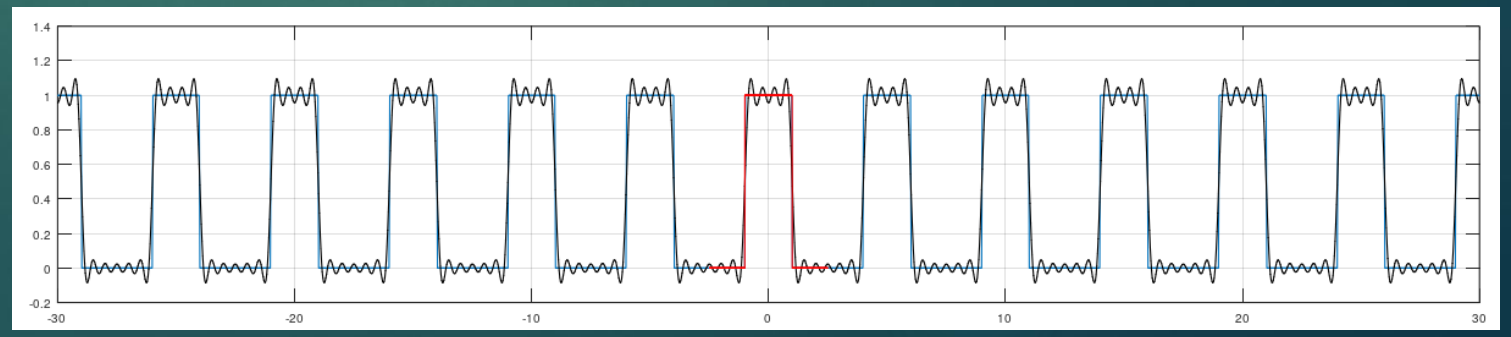
No limite, se tivermos infinitos coeficientes da série F(n), $com\ \ -\infty < n < \infty$, podemos reconstituir exatamente a função periódica $f(t)$ através da expressão $f(t) = \sum_{n=-\infty}^{\infty} F(n)e^{jnw_o t}$

No entanto, se a série for truncada (como no cálculo acima, com 10 termos positivos), teremos um reconstituição aproximada da função periódica $f(t)$, que será tanto melhor quantos coeficientes forem utilizados

$$f(t) \approx \sum_{n=-9}^{9} F[n]e^{jnw_o t} = \sum_{n=-9}^{9} F[n]e^{jn\left(\frac{2\pi}{5}\right)t}$$

```
>> y=SinteseSerieFourier(Fn,tp,-2.5,2.5);
>> plot(tp,y,'k');
```

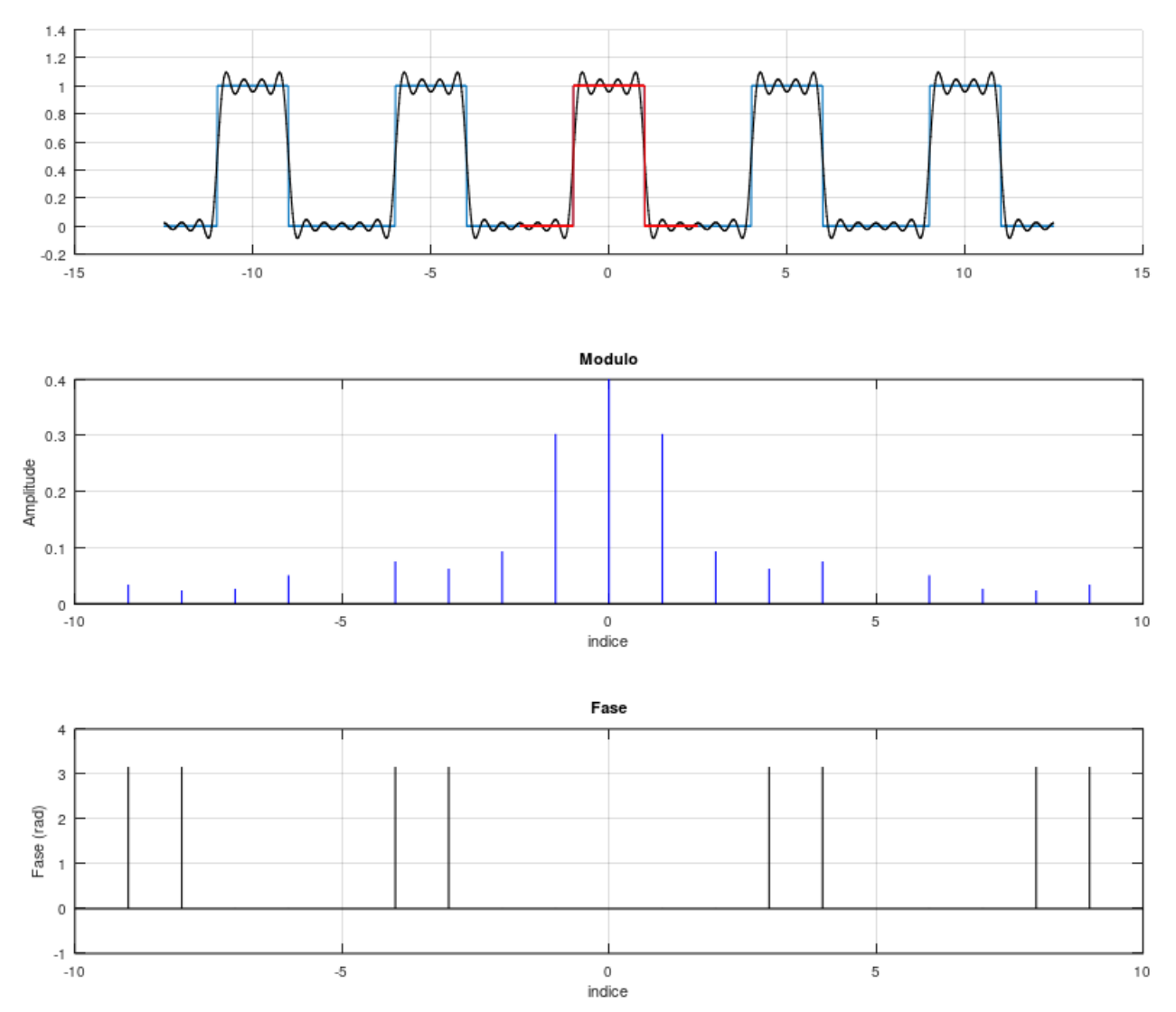
```
function y=SinteseSerieFourier(F,t,t_ini,t_fim)
    T=(t_fim-t_ini);
    w0 = 2* pi  /T;
    N=(length(F)+1)/2;
    y=F*exp( i *[-N+1:N-1]'*w0*t);
endfunction
```



Função que plota 5 períodos da função periódica $f(t)$, calcula e plota os N primeiros coeficientes da série de Fourier $F(n)$, e reconstitui (parcialmente) $f(t)$ com os N coeficientes.

```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> PlotSerieFourier(f2,-2.5,2.5,10)
```

```
function PlotSerieFourier(f,t_ini,t_fim,N)
    clf;
    F=SerieFourier(f,t_ini,t_fim,N);
    T=(t_fim-t_ini);
    w0 = 2* pi  /T;
   subplot(3,1,1)
    t3=linspace(t_ini-2*T,t_fim+2*T,2000);
    grid on; hold on;
    plot(t3,MakePeriodic(f,t3,t_ini,t_fim));
    y3=SinteseSerieFourier(F,t3,t_ini,t_fim); %3 periodos
    plot(t3,y3,'bk');
    t1=linspace(t_ini,t_fim,1000);
    plot(t1,f(t1),'r');  %  1 periodo
   subplot(3,1,2)
    n=[-N+1:N-1];
    stem(n,abs(F),'b','marker','none'); % modulo
    axis([-N,N]);
    grid on;
    title("Modulo"); xlabel("indice"); ylabel("Amplitude");
   subplot(3,1,3)
    fase = atan2(imag(F),real(F));
    index = find((abs(F)<max(abs(F)/1000))|(abs(fase)<1e-4));
    fase(index)=0;
    index = find(fase<=-3);
    fase(index)=fase(index)+2* pi;
    stem(n,fase,'k','marker','none'); % fase
    axis([-N,N]);
    grid on;
    title("Fase");xlabel("indice");ylabel("Fase (rad)");
endfunction
```

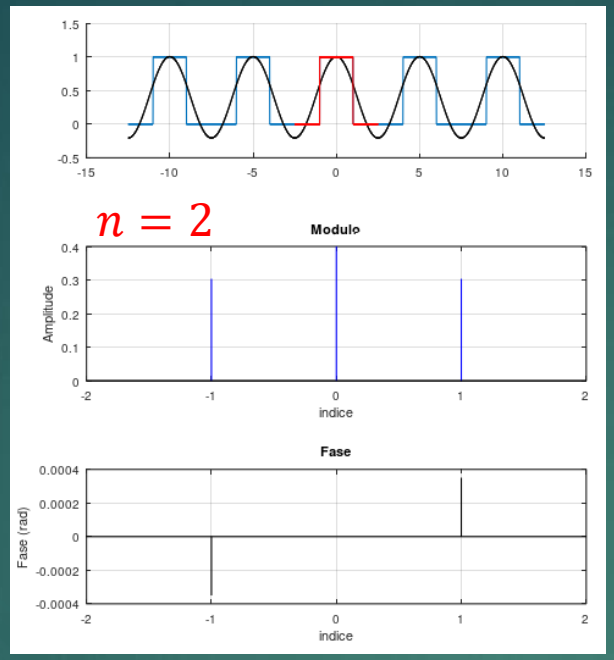


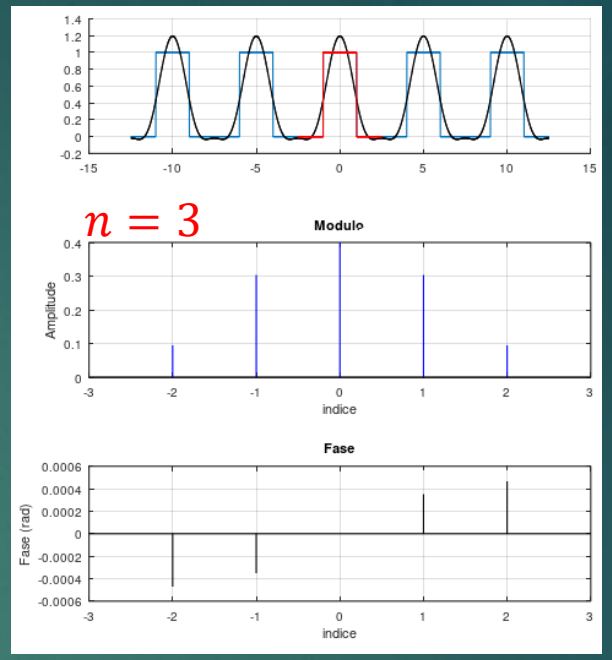
```
>> f2 = @(t) f_u(t+1)-f_u(t-1);
>> PlotSerieFourier(f2,-2.5,2.5,2)
>> PlotSerieFourier(f2,-2.5,2.5,3)
>> PlotSerieFourier(f2,-2.5,2.5,4)
>> PlotSerieFourier(f2,-2.5,2.5,5)
>> PlotSerieFourier(f2,-2.5,2.5,10)
>> PlotSerieFourier(f2,-2.5,2.5,100)
```

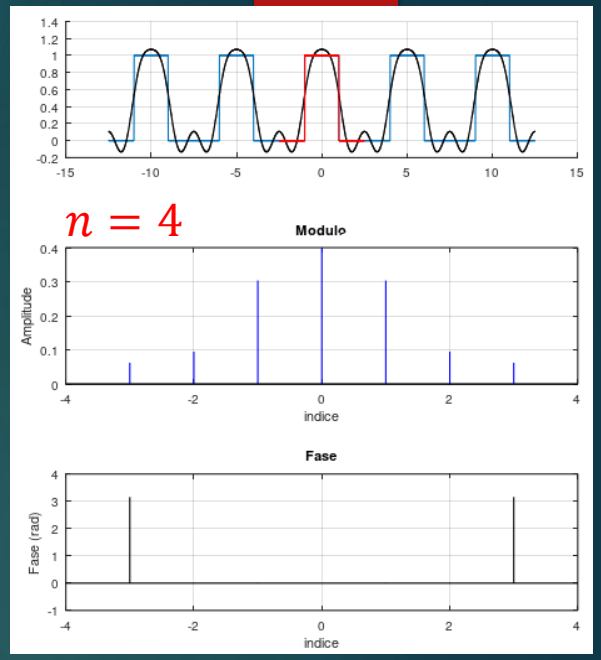
Notar que com $n = 2$ só temos energia em $n = -1\ e\ n = 1$; e um nível DC (n=0).

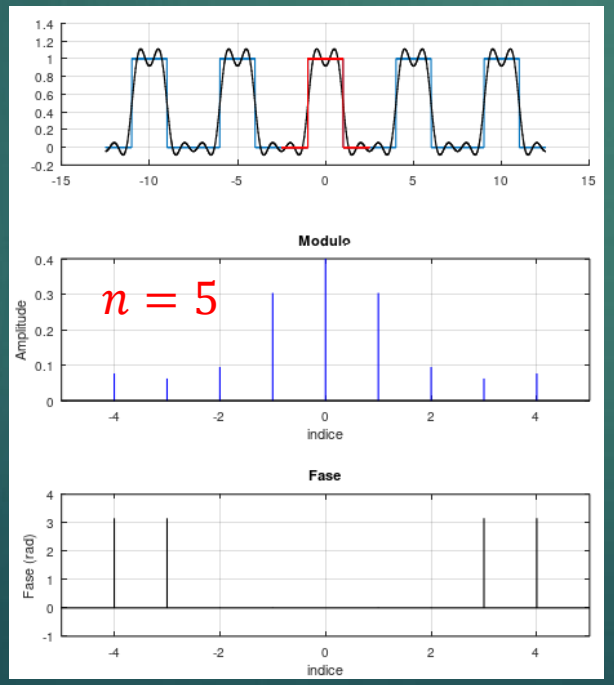
Quando aumentamos o número de harmônicos, a reconstituição fica mais aproximada.

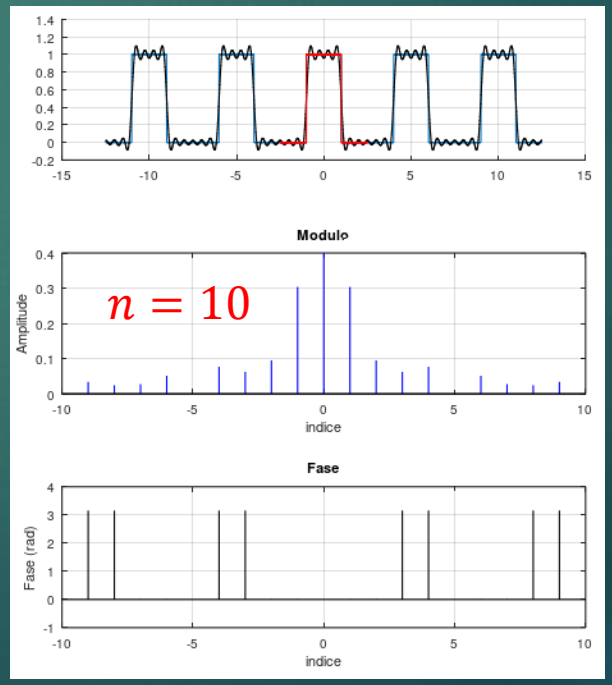
No entanto, mesmo com $n = 100$, ainda temos o Efeito Gibbs nas descontinuidades.

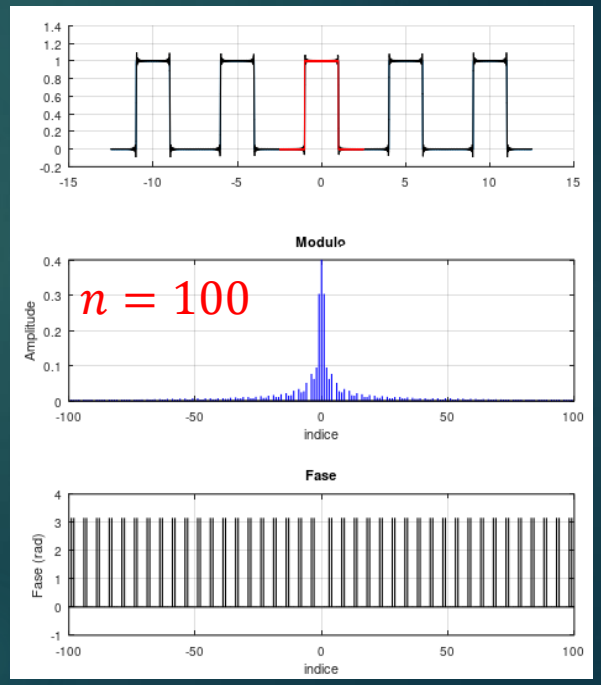












Exemplo 2 - Onda Quadrada

$$f(t) = \begin{cases} -1; & -\pi \le t < 0 \\ 1; & 0 \le t < \pi \end{cases} \quad ; \quad f(t+2\pi) = f(t)$$

Notar que f(t) é periódica com período 2π

$$T = 2\pi \qquad w_o = \frac{2\pi}{T} = 1$$

Cálculo analítico da Série:

$$F[n] = \frac{1}{T}\int_0^T f(t)e^{(-jnw_o t)}dt = \frac{1}{2\pi}\left[\int_{-\pi}^{0}(-1)e^{(-jn\,1\,t)}dt + \int_0^{\pi}(1)e^{(-jn\,1\,t)}dt\right] = \begin{cases} 0; & se\ n\ par \\ \dfrac{-2j}{n\pi}; & se\ n\ ímpar \end{cases}$$

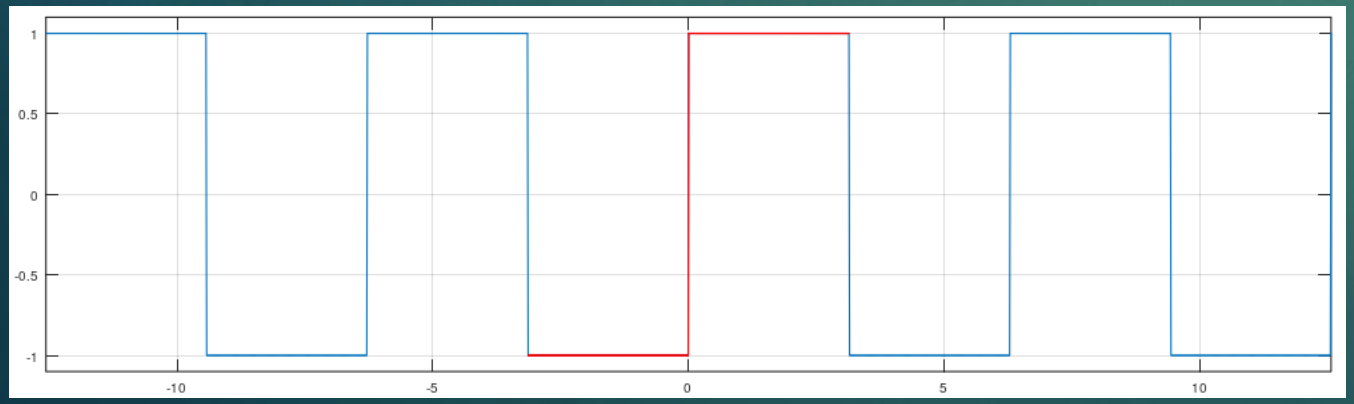
```
>> f4 = @(t) 2*f_u(t)-1;
>> f_T = @ (t) MakePeriodic(f4,t,-pi,pi);
>> tp=linspace(-4*pi,4*pi,2000);
>> plot(tp,f_T(tp));
>> hold on; grid on;
>> t=linspace(-pi,pi,1001);
>> plot(t,f4(t),'r');
```

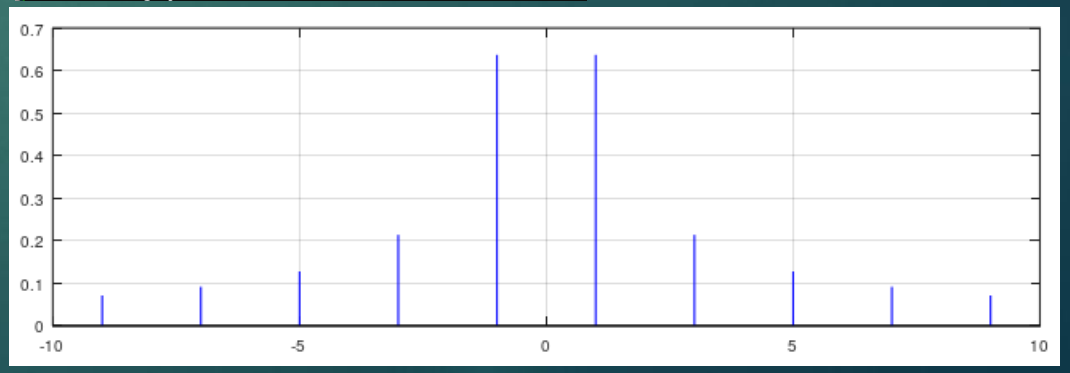
```
function C = F4(n)
    C = -2*i./(n*pi);
    index=find(mod(n,2)==0);
    C(index)=0;
endfunction;
```

```
>> n=[-9:9];
>> [n' F4(n)']
ans =

  -9   0 - 0.0707i
  -8   0 -      0i
  -7   0 - 0.0909i
  -6   0 -      0i
  -5   0 - 0.1273i
  -4   0 -      0i
  -3   0 - 0.2122i
  -2   0 -      0i
  -1   0 - 0.6366i
       0 -      0i
   1   0 + 0.6366i
   2   0 -      0i
   3   0 + 0.2122i
   4   0 -      0i
   5   0 + 0.1273i
   6   0 -      0i
   7   0 + 0.0909i
   8   0 -      0i
   9   0 + 0.0707i
```

```
>> n=[-9:9];
>> bar(n,abs(F4(n)),'b',0.2)
```





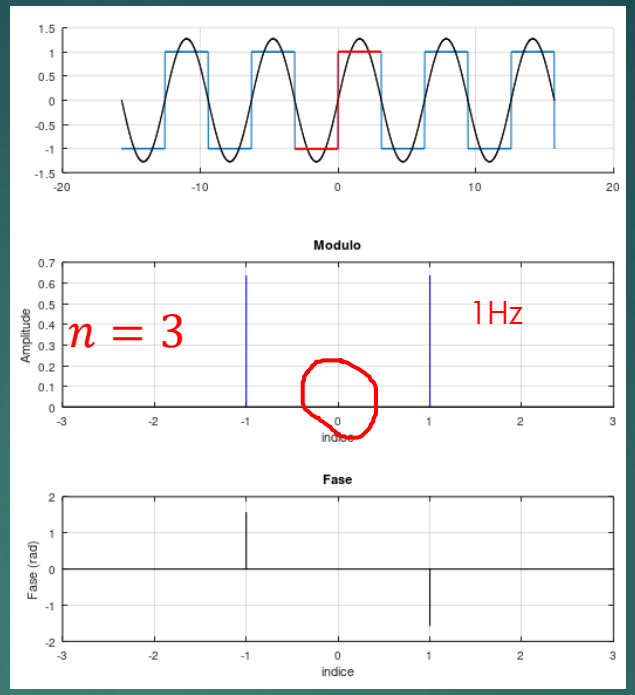
```
>> f4 = @(t) 2*f_u(t)-1;
>> PlotSerieFourier(f4,-pi,pi,3)
>> PlotSerieFourier(f4,-pi,pi,5)
>> PlotSerieFourier(f4,-pi,pi,11)
>> PlotSerieFourier(f4,-pi,pi,101)
```

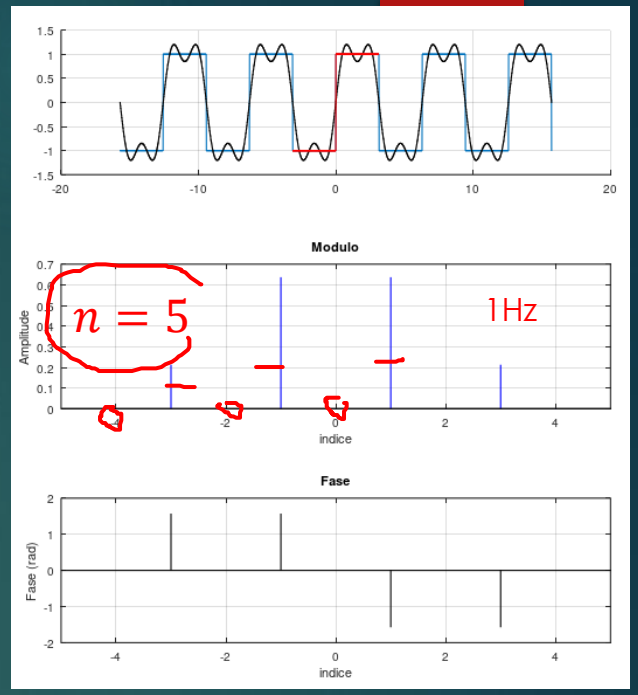
Notar que com $n = 3$ só temos energia em $n = -1$ e $n = 1$; e não há nível DC.

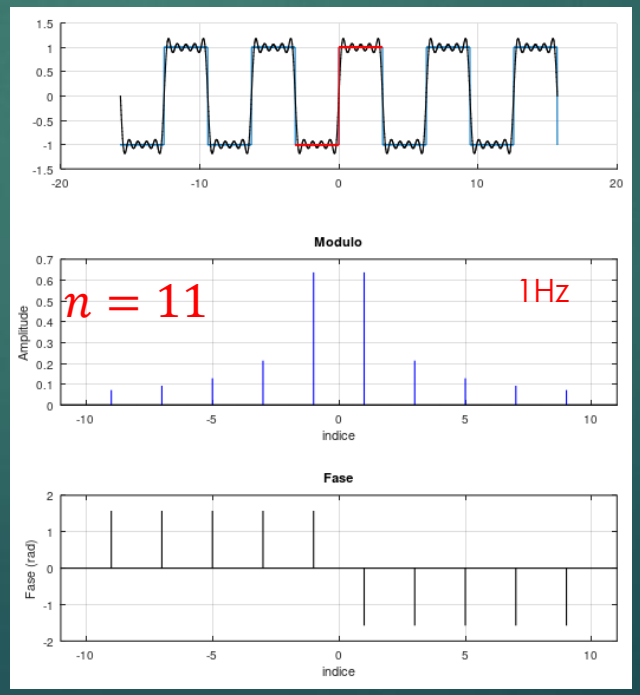
Notar que os harmônicos pares são nulos, pois $f(t)$ é impar!.

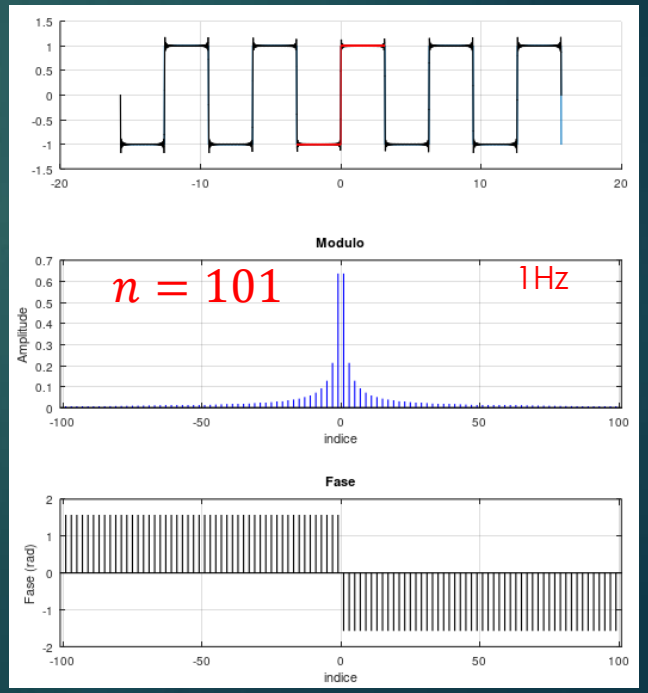
Quando aumentamos o número de harmônicos, a reconstituição fica mais aproximada.

No entanto, mesmo com $n = 100$, ainda temos o Efeito Gibbs nas descontinuidades.









Exemplo 3 - Onda Triangular

$f(t) = |t|\ ;\ -10 \le t \le 10;\quad f(t+20) = f(t)$

Notar que f(t) é periódica com período 20

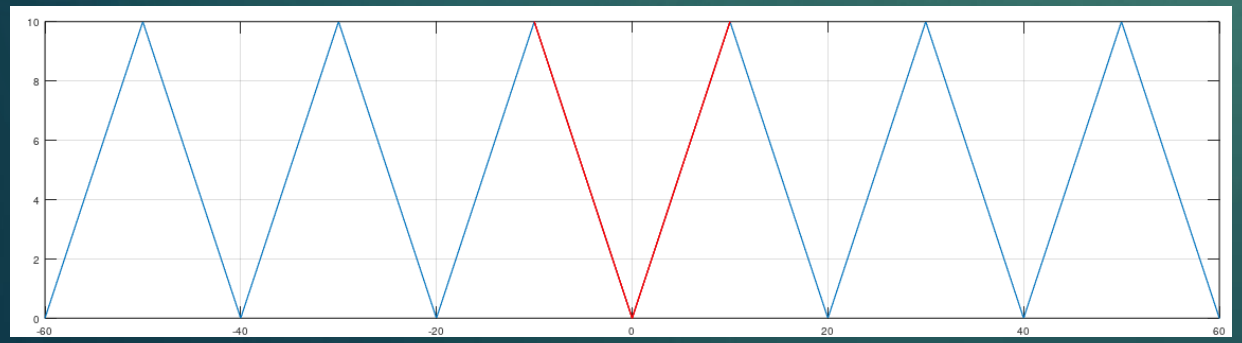
$T = 20$

$w_o = \frac{2\pi}{20} = \frac{\pi}{10}$

Cálculo analítico da Série:

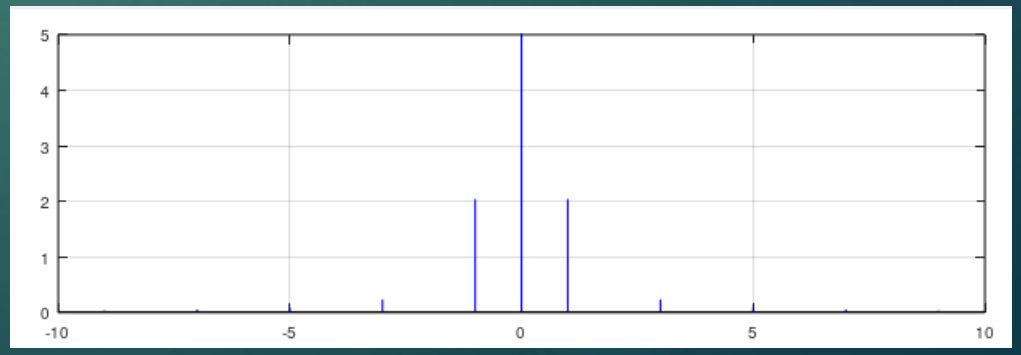
$$F[n] = \frac{1}{T}\int_0^T f(t)e^{(-jnw_o t)}dt = \frac{1}{20}\left[\int_{-10}^{0}\left(-\frac{t}{2}\right)e^{\left(jn\frac{\pi}{10}t\right)}dt + \int_0^{10}\left(\frac{t}{2}\right)e^{\left(jn\frac{\pi}{10}t\right)}dt\right] = \begin{cases} 5; & n = 0 \\ 0; & n\ par \\ -\dfrac{20}{n^2\pi^2}; & n\ ímpar \end{cases}$$

```
>> f5 = @(t) abs(t);
>> f_T = @ (t) MakePeriodic(f5,t,-10,10);
>> tp=linspace(-60,60,2000);
>> plot(tp,f_T(tp));
>> t=linspace(-10,10,2000);
>> hold on; grid on;
>> plot(t,f5(t),'r');
```




```
function C = F5(n)
    C = n*0;
    index=find(mod(n,2)==1);
    C(index)=-20./(n(index).^2*pi^2);
    index=find(n==0);
    C(index)=5;
endfunction;
```

```
>> n=[-9:9];
>> stem(n,abs(F5(n)),'b','marker','none');
```




```
>> n=[-9:9];
>> [n' F5(n)']
ans =

   -9.0000   -0.0250
   -8.0000         0
   -7.0000   -0.0414
   -6.0000         0
   -5.0000   -0.0811
   -4.0000         0
   -3.0000   -0.2252
   -2.0000         0
   -1.0000   -2.0264
         0    5.0000
    1.0000   -2.0264
    2.0000         0
    3.0000   -0.2252
    4.0000         0
    5.0000   -0.0811
    6.0000         0
    7.0000   -0.0414
    8.0000         0
    9.0000   -0.0250
```

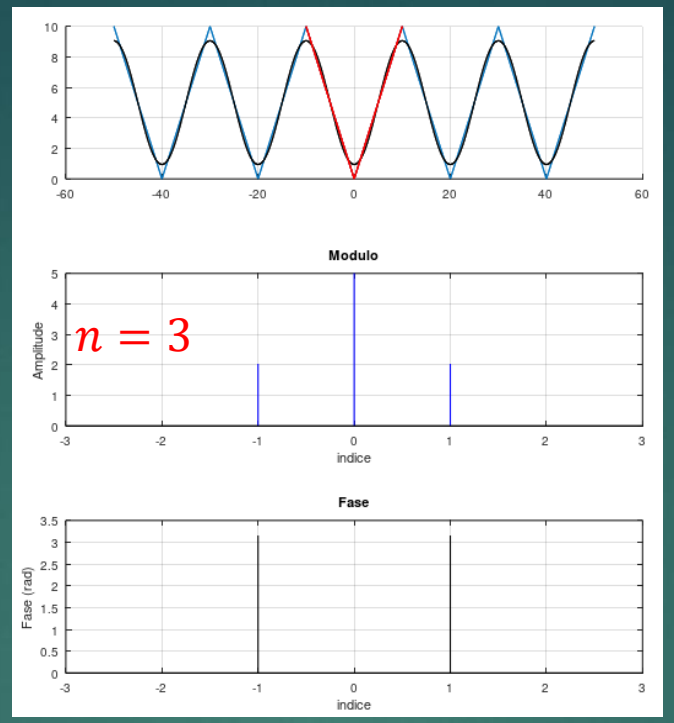
```
>> f5 = @(t) abs(t);
>> PlotSerieFourier(f5,-10,10,3)
>> PlotSerieFourier(f5,-10,10,5)
>> PlotSerieFourier(f5,-10,10,11)
>> PlotSerieFourier(f5,-10,10,101)
```

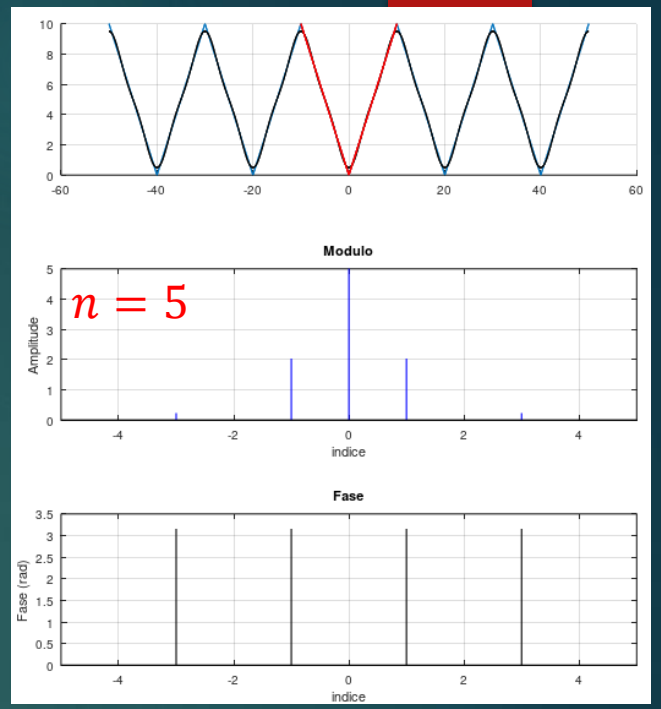
Notar que com $n = 3$ só temos energia em $n = -1$ e $n = 1$; e um nível DC (n=0).

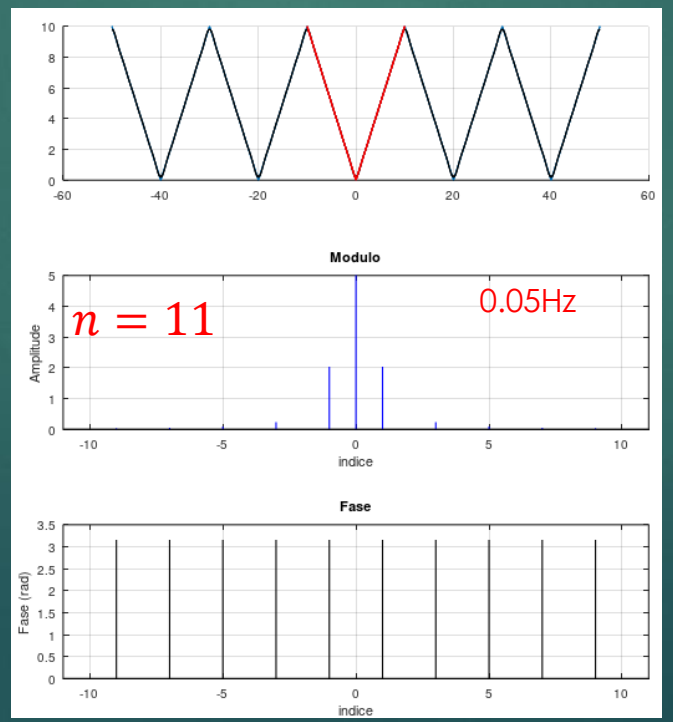
Notar que os harmônicos pares são nulos, pois $f(t)$ é impar!.

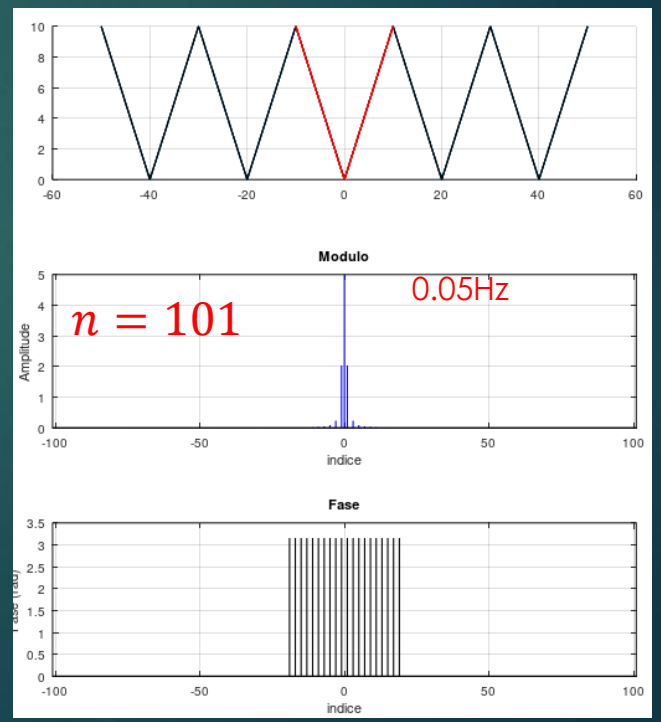
Quando aumentamos o número de harmônicos, a reconstituição fica mais aproximada.

Como não há descontinuidades em $f(t)$, com $n = 100$ a reconstituição é muito satisfatória (não há efeito Gibs)









Exemplo 4 - $f(t) = A\cos\left(\frac{2\pi}{7}t\right)$; $w_o = \frac{2\pi}{7}$; $T = \frac{w_0}{2\pi} = 7$; 1^0 Período $0 \leq t \leq 7$

```
>> w0=2*pi/7;
>> f6 = @(t) 10*cos(w0*t);;
>> PlotSerieFourier(f6,0,7,10)
```

$$F[n] = \frac{a_n}{2} \quad (par)$$

$$F[0] = a_o = \frac{1}{T}\int_0^T f(t)dt = \frac{1}{T}\int_0^T A\cos(w_o t)dt = 0$$

$$a_n = \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\cos(nw_o t)\,dt = \frac{8\pi}{w_o}\int_0^{\frac{T}{2}}[A\cos(w_o t)]\cos(nw_o t)\,dt$$

$se\ n \neq 1$

$$a_n = \int_0^{\frac{T}{2}}[A\cos(w_o t)]\cos(nw_o t)\,dt = 0$$

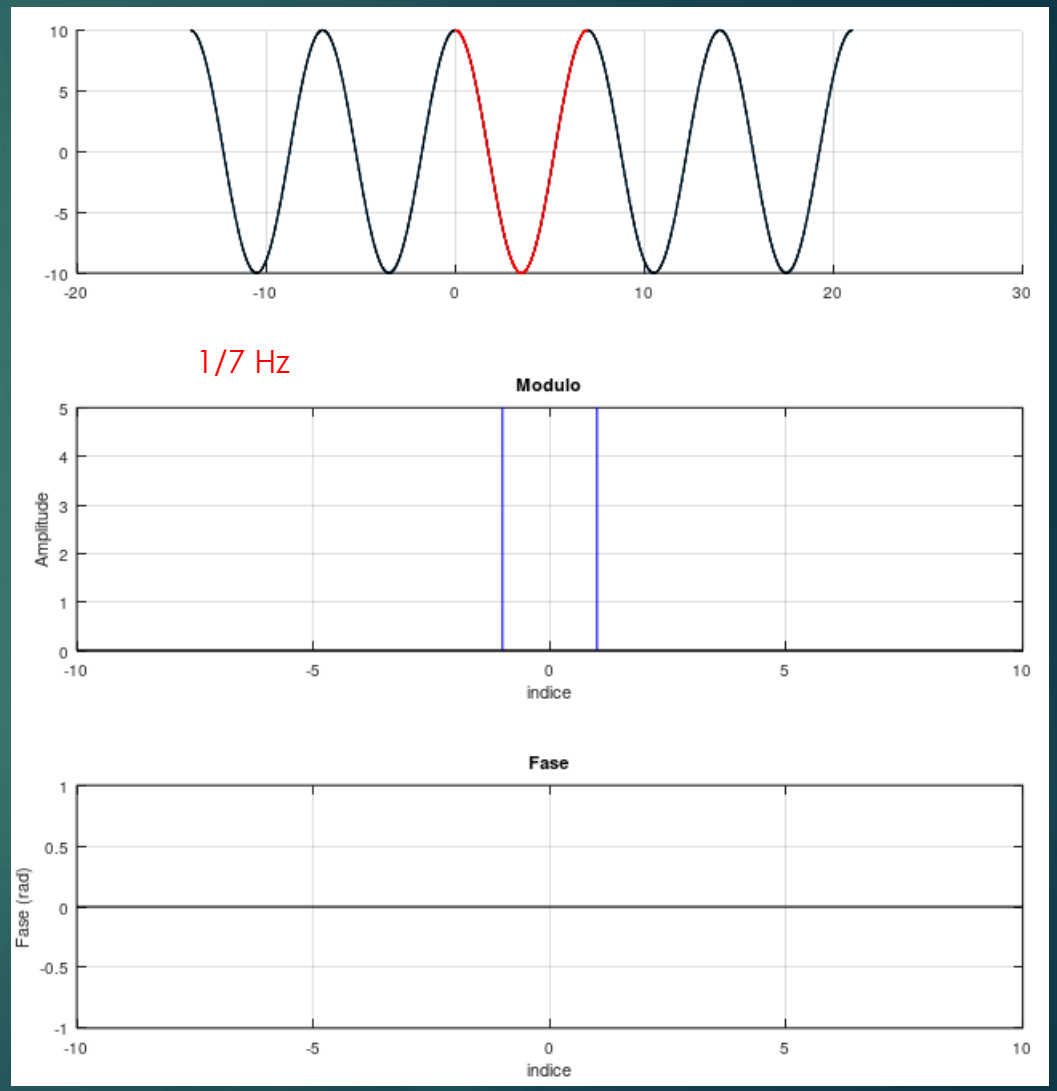
$se\ n = 1$

$$a_1 = \frac{8\pi}{w_o}\int_0^{\frac{T}{2}}[A\cos(w_o t)]\cos(1w_o t)\,dt = \frac{4A}{T}\left(\frac{T}{4}\right) = A$$

$$F[n] = \left[\ldots 0,0,0,0,\frac{A}{2},0,\frac{A}{2},0,0,0,0\ldots\right]$$

Como f(t) tem somente uma frequência $w_o = \frac{2\pi}{7}$ com fase zero (cosseno) e amplitude A, a sua Série de Fourier $F(n)$ vai conter somente energia em $n = -1\ e\ n = 1$ com amplitude A/2.

Notar que em $n = -1\ e\ n = 1$ a fase é nula.



Exemplo 5

$$f(t) = B\ sen\left(\frac{2\pi}{7}t\right); \quad w_o = \frac{2\pi}{7}; \quad T = \frac{w_0}{2\pi} = 7; \quad 1^0 \text{ Período } 0 \le t \le 7$$

```
>> w0=2*pi/7;
>> f7 = @(t) 10*sin(w0*t);;
>> PlotSerieFourier(f7,0,7,10)
```

$$F[n] = -j\frac{b_n}{2} \quad (\text{ímpar})$$

$$b_n = \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\,\text{sen}(nw_o t)\,dt = \frac{4}{T}\int_0^{\frac{T}{2}} [B\ sen(w_o t)]\,\text{sen}(nw_o t)\,dt$$

$se\ n \neq 1$

$$b_n = \frac{4}{T}\int_0^{\frac{T}{2}} [B\ sen(w_o t)]\,\text{sen}(nw_o t)\,dt = 0$$

$se\ n = 1$

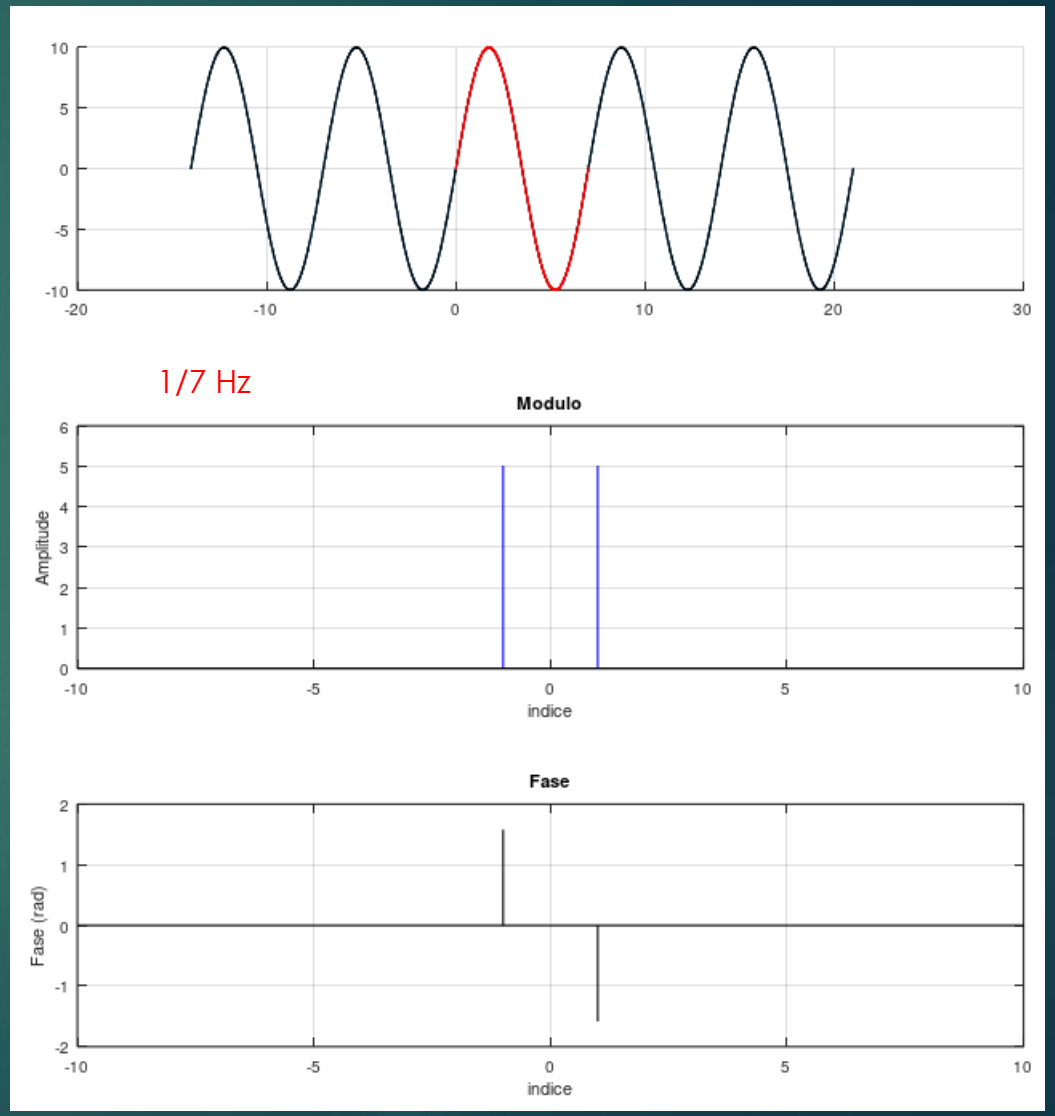
$$b_1 = \frac{4}{T}\int_0^{\frac{T}{2}} [B\ sen(w_o t)]\,\text{sen}(1w_o t)\,dt = \frac{4B}{T}\left(\frac{T}{4}\right) = B$$

$$F[n] = \left[\ldots 0,0,0,0,j\frac{B}{2},0,-j\frac{B}{2},0,0,0,0\ldots\right]$$

Notar que f(t) tem somente uma frequência $w_o = \frac{2\pi}{7}$ com fase $\pi/2$ (pois é um seno) e amplitude A.

A sua Série de Fourier $F(n)$ vai conter somente energia em $n = -1$ e $n = 1$ com amplitude A/2.

Notar que em $n = -1\ e\ n = 1$ a fase é $\pi/2$ e $-\pi/2$ respectivamente.



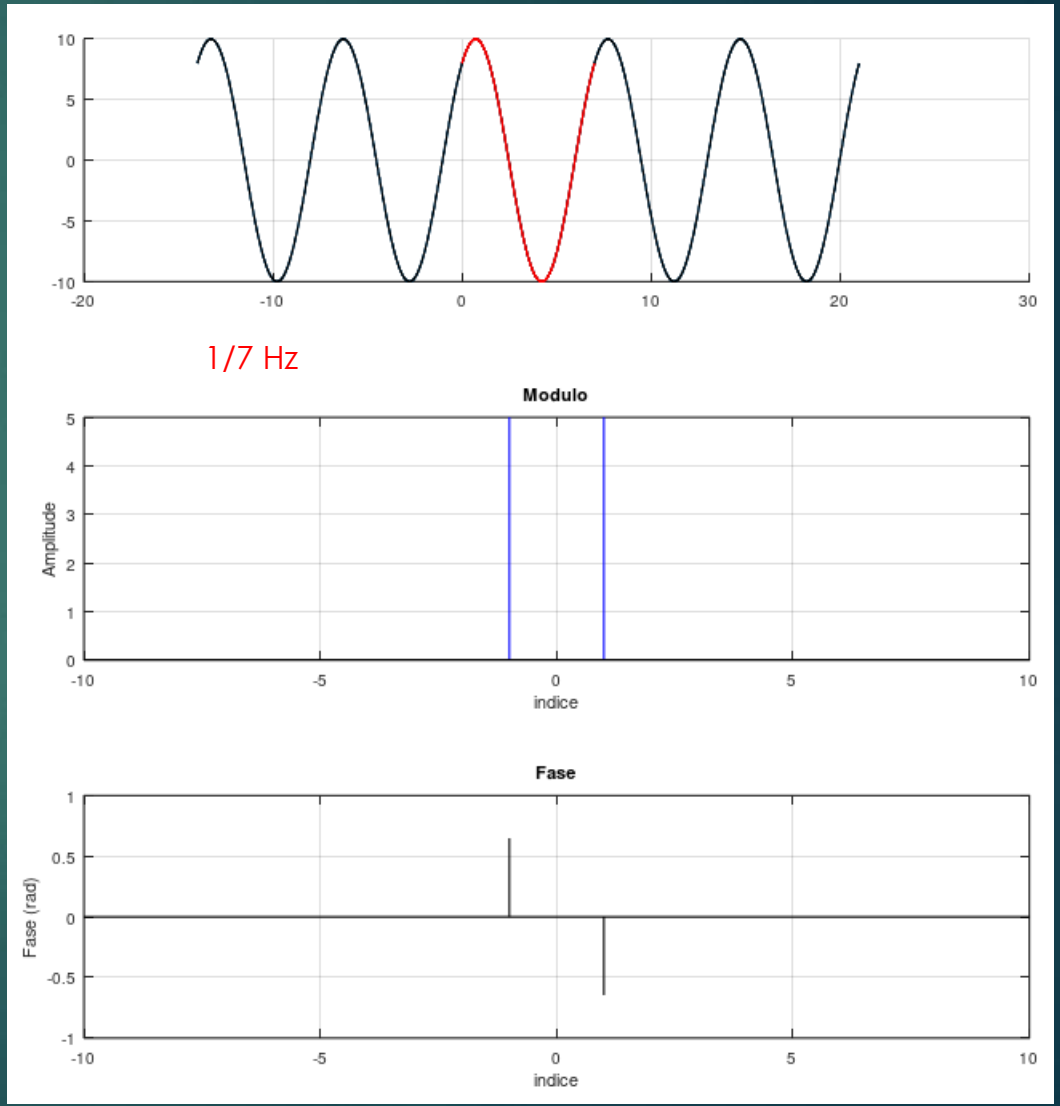
Exemplo 6-

$$f(t) = 6\,sen\left(\frac{2\pi}{7}t\right) + 8\cos\left(\frac{2\pi}{7}t\right); \quad w_o = \frac{2\pi}{7}; \quad T = \frac{w_0}{2\pi} = 7; \quad 1^0 \text{ Período } 0 \le t \le 7$$

```
>> w0=2*pi/7;
>> f8 = @(t) 6*sin(w0*t)+8*cos(w0*t);
>> PlotSerieFourier(f8,0,7,10)
```

$$\mathrm{F}[n] = \frac{a_n}{2} - j\frac{\mathrm{b_n}}{2} \qquad F(-n) = F^*(n)$$

$$\mathrm{F}[n] = \left[\ldots 0,0,0,0,\frac{4}{2} + j\frac{6}{2}, 0, \frac{5}{2} - j\frac{6}{2}, 0, 0, 0, 0 \ldots\right]$$



Notar que f(t) tem somente uma frequência $\mathrm{w_o} = \frac{2\pi}{7}$ com fase $\mathrm{atan}\left(\frac{A_1}{A_2}\right)$, pois é uma combinação de um cosseno com amplitude A_2 e um seno com amplitude A_1.

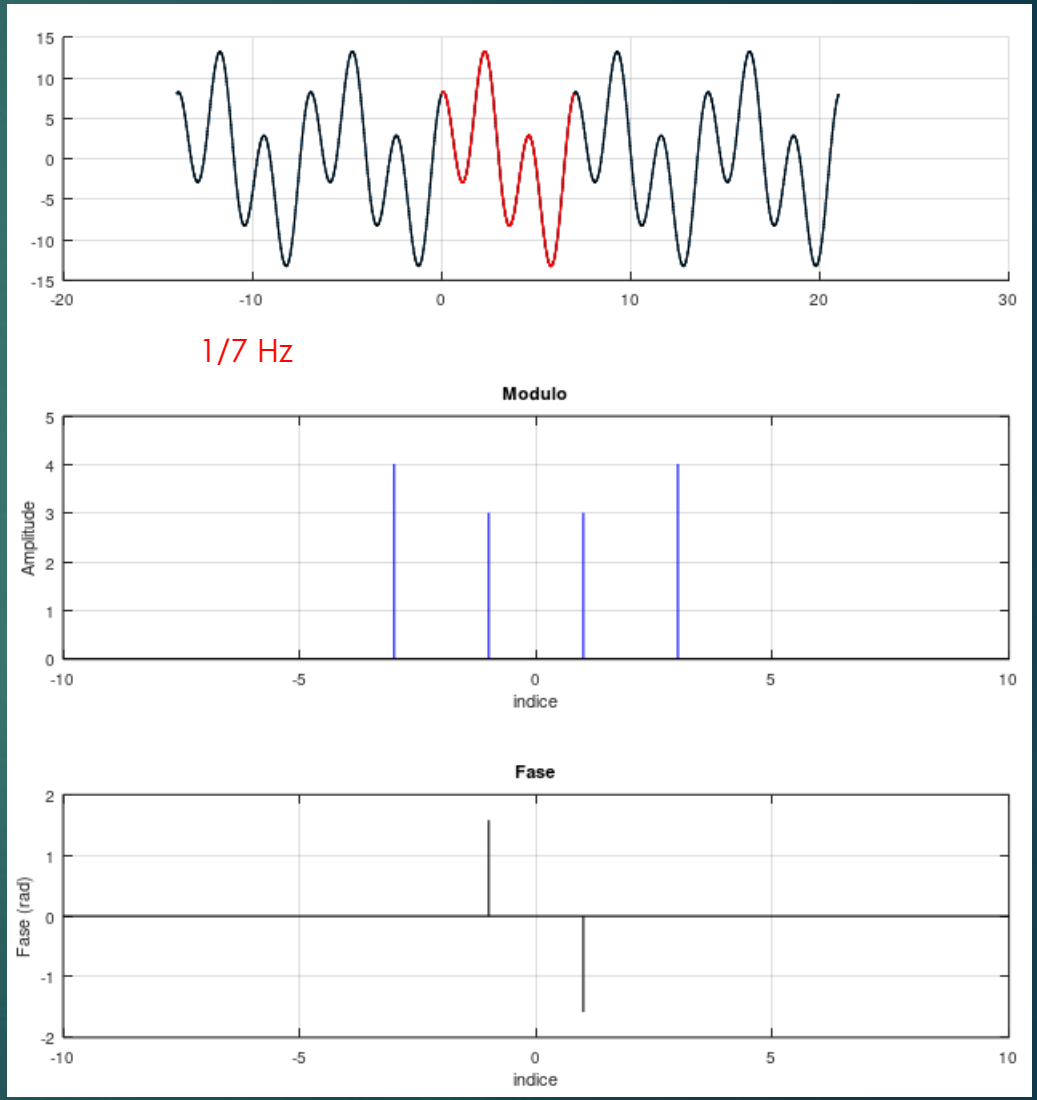
Q sua Série de Fourier $F(n)$ vai conter somente energia em $n = -1\ e\ n = 1$ com amplitude $\sqrt{A_1^2 + A_2^2}$.

Notar que em $n = -1\ e\ n = 1$ a fase é $\mathrm{atan}\left(\frac{A_1}{A_2}\right)$ e $\mathrm{atan}\left(\frac{A_1}{A_2}\right)$ respectivamente.

Exemplo 7-

$f(t) = 6\,sen(w_o t) + 8\cos(3w_o t)\,;\quad w_o = \frac{2\pi}{7};\quad T = \frac{w_0}{2\pi} = 7;\quad 1^0$ Período $0 \le t \le 7$

```
>> w0=2*pi/7;
>> f9 = @(t) 6*sin(w0*t)+8*cos(3*w0*t);
>> PlotSerieFourier(f9,0,7,10)
```



Notar que f(t) tem duas frequências: $w_1 = \frac{2\pi}{7}$ com fase $\pi/2$ (seno) e amplitude 6, e $w_2 = 3\frac{2\pi}{7}$ com fase *zero* (cosseno) e amplitude 8.

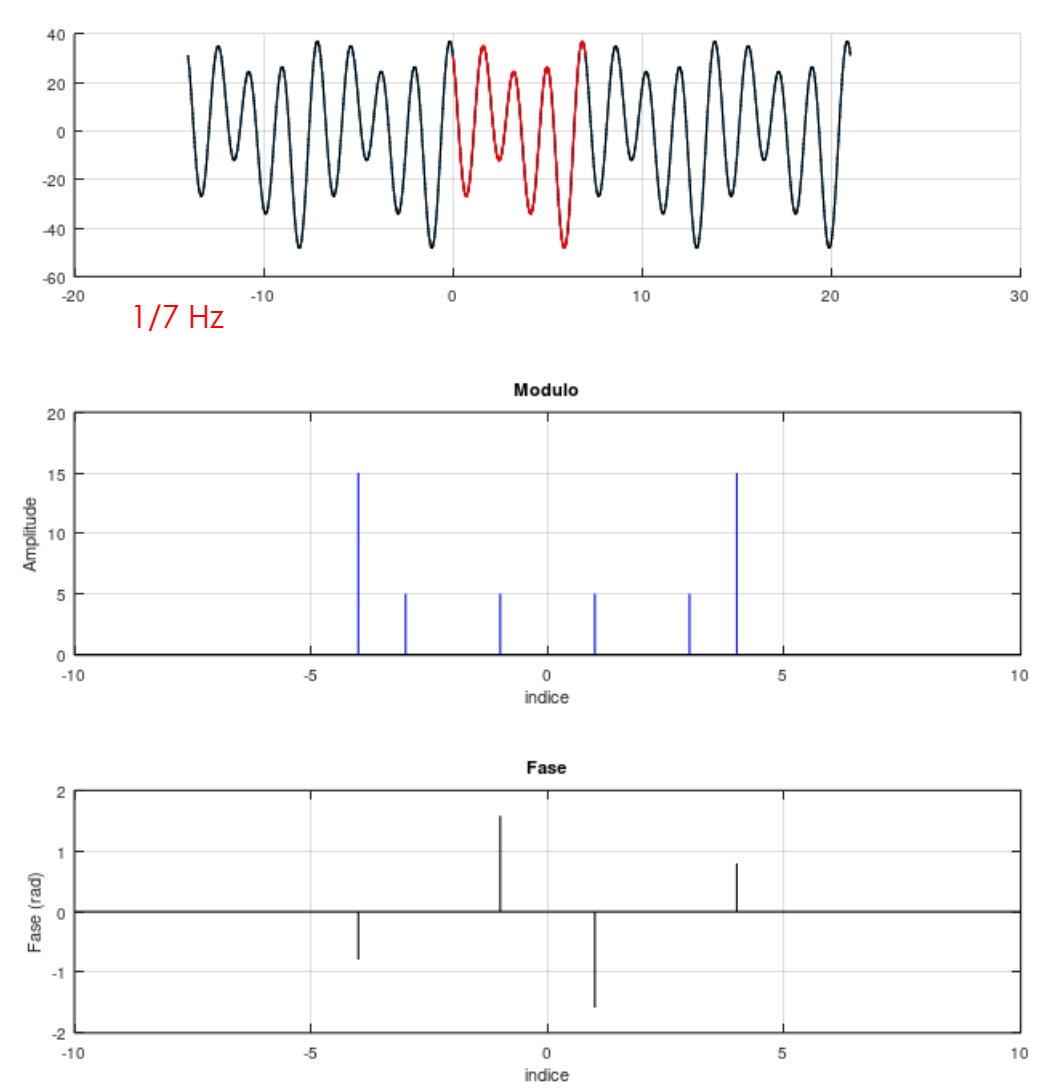
Notar que período T de repetição é definido pela menor frequência w_o (e não por $3w_o$)

A sua Série de Fourier $F(n)$ vai conter energia em $n = -1\ e\ n = 1$ com amplitude 3 e fase $\pi/2$ e $-\frac{\pi}{2}$; energia em $n = -3\ e\ n = 3$ com amplitude 4 e fase zero

Exemplo 8-

$f(t) = 10\,sin(w_o t) + 10\cos(3w_o t) + 30cos(4w_o t + \frac{\pi}{4})$; $w_o = \frac{2\pi}{7}$; $T = \frac{w_0}{2\pi} = 7$; 1^0 Período $0 \leq t \leq 7$

```
>> w0=2*pi/7;
>> f10 = @(t) 10*sin(w0*t)+10*cos(3*w0*t)+30*cos(4*w0*t+pi/4);
>> PlotSerieFourier(f10,0,7,10)
```



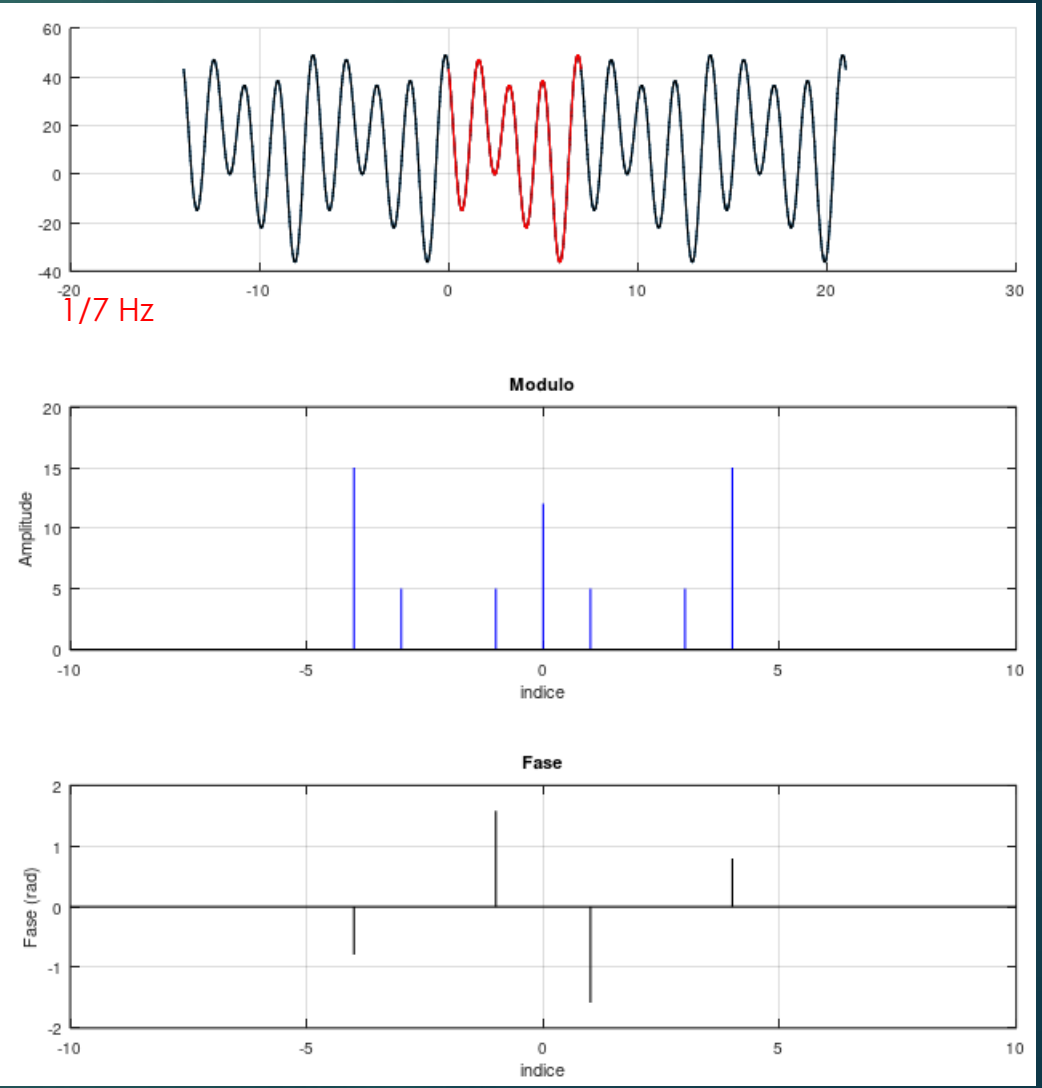
Notar que f(t) tem três frequências : $w_1 = \frac{2\pi}{7}$ com fase $\pi/2$ (seno) e amplitude 10, e $w_2 = 3\frac{2\pi}{7}$ com fase *zero* (cosseno) e amplitude 10; e $w_3 = 4\frac{2\pi}{7}$ com fase $\pi/4$ e amplitude 30

Notar que período T de repetição é definido pela menor frequência w_o (e não por $3w_o$ ou $4w_o$)

A sua Série de Fourier $F(n)$ vai conter energia em $n = -1\ e\ n = 1$ com amplitude 5 e fase $\pi/2$ e $-\pi/2$; e energia em $n = -3\ e\ n = 3$ com amplitude 5 e fase zero; e energia em $n = -4\ e\ n = 4$ com amplitude 15 fase $-\pi/4$ e $\pi/4$.

Exemplo 9- $f(t) = 12 + 10\,sin(w_o t) + 10\cos(3w_o t) + 30cos(4w_o t + \frac{\pi}{4})$; $w_o = \frac{2\pi}{7}$; $T = \frac{w_0}{2\pi} = 7$; 1^0 Período $0 \leq t \leq 7$

```
>> w0=2*pi/7;
>> f11 = @(t) 12+10*sin(w0*t)+10*cos(3*w0*t)+30*cos(4*w0*t+pi/4);
>> PlotSerieFourier(f11,0,7,10)
```



Notar que f(t) tem três frequências : $w_1 = \frac{2\pi}{7}$ com fase $\pi/2$ (seno) e amplitude 10, e $w_2 = 3\frac{2\pi}{7}$ com fase *zero* (cosseno) e amplitude 10; e $w_3 = 4\frac{2\pi}{7}$ com fase $\pi/4$ e amplitude 30; e um nível DC de 12;

Notar também que período T de repetição é definido pela menor frequência não nula w_o (e não por $3w_o$ ou $4w_o$).

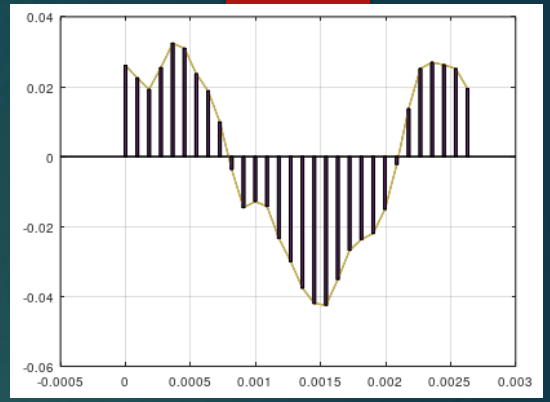
A sua Série de Fourier $F(n)$ vai conter energia em $n = -1\ e\ n = 1$ com amplitude 5 e fase $\pi/2$ e $-\pi/2$; e energia em $n = -3\ e\ n = 3$ com amplitude 5 e fase zero; e energia em $n = -4\ e\ n = 4$ com amplitude 15 fase $-\pi/4$ e $\pi/4$; e energia em $n = 0$ com amplitude 12;

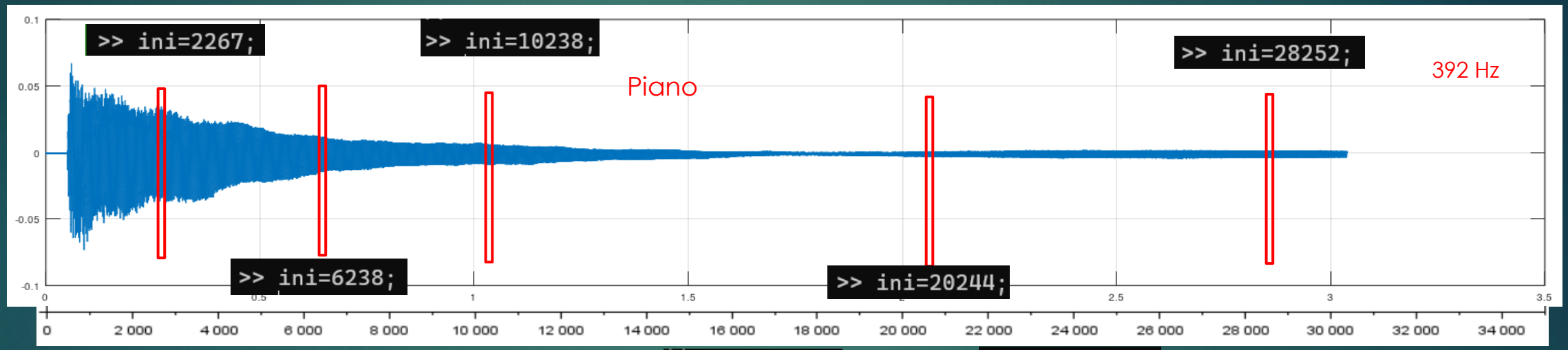
Exemplo 10 - Instrumentos Musicais- Séries Harmônicas e Timbres

```
>> [y,fs]=audioread("./Dados/piano.wav");
>> Ts=1/fs
Ts = 9.0703e-05
>> N=length(y)
N = 33500
>> t=[0:N-1]*Ts;
>> plot(t,y)
```

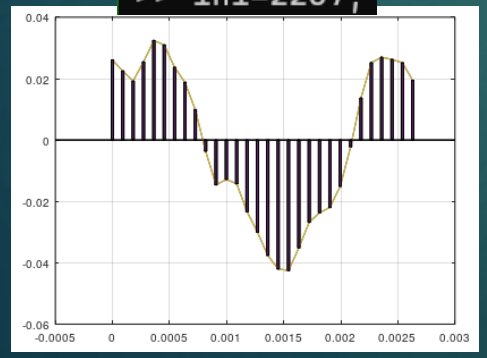
```
>> f0=392;
>> Np=ceil(fs/f0)
Np = 29
>> tp=[0:Np]*Ts;
```

```
>> ini=2267;
>> plot(tp,y(ini:ini+Np)-mean(y(ini:ini+Np)));
>> hold on; grid on;
>> bar(tp,y(ini:ini+Np)-mean(y(ini:ini+Np)),0.2);
```

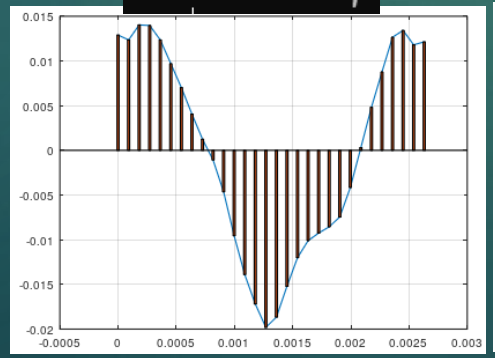




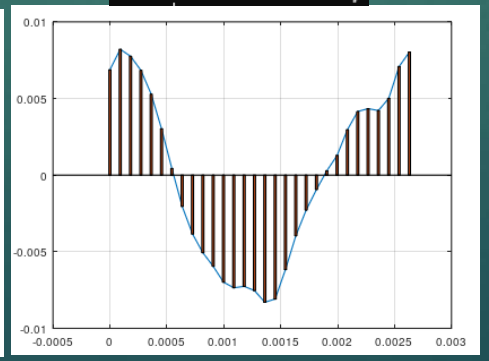
>> ini=2267;



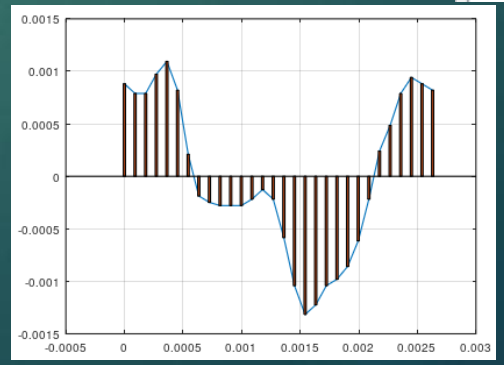
>> ini=6238;



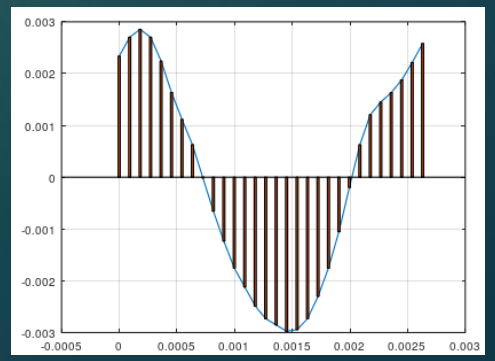
>> ini=10238;

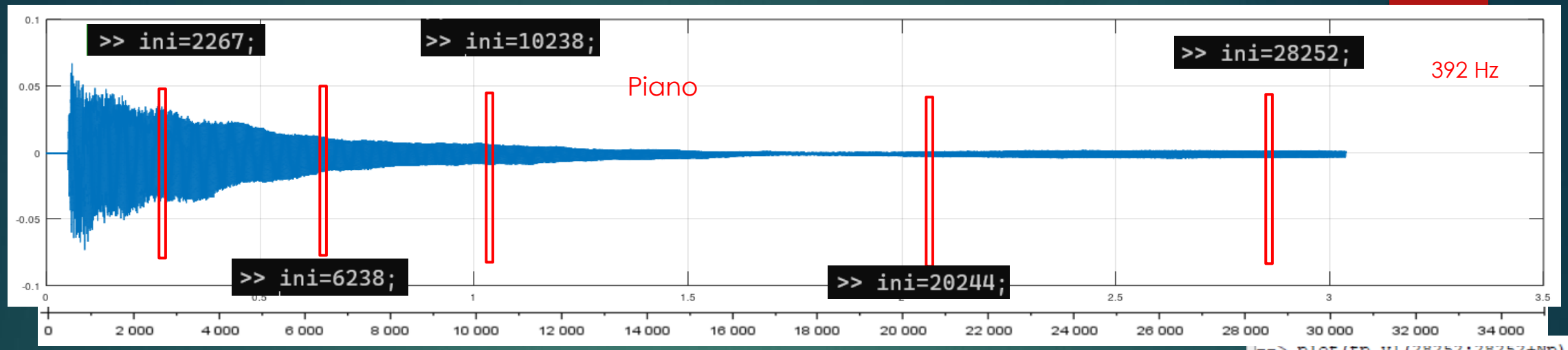


>> ini=20244;

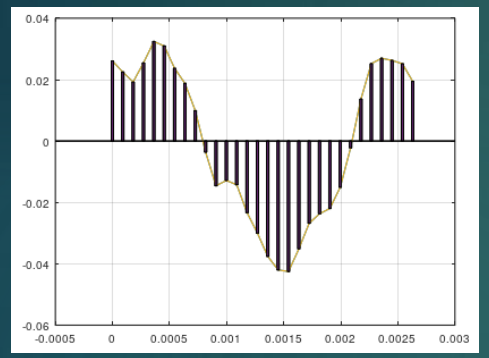


--> plo >> ini=28252; 2+Np))

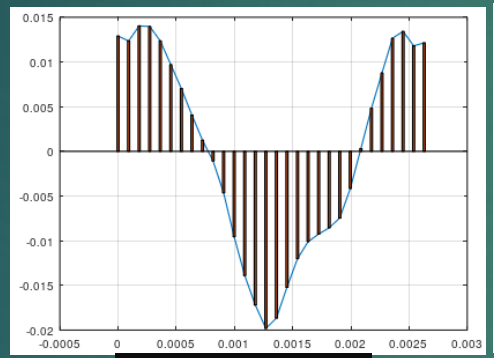




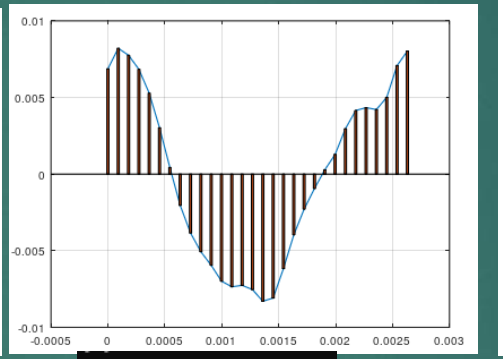

```
|--> plot(tp,y1(28252:28252+Np))
```



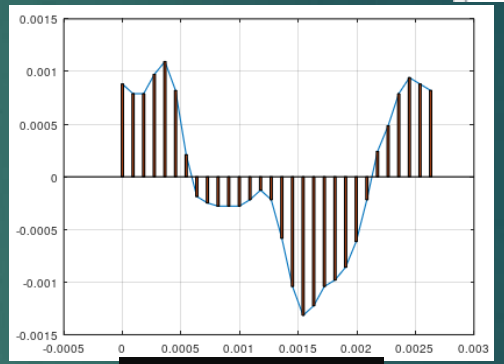
>> ini=2267;



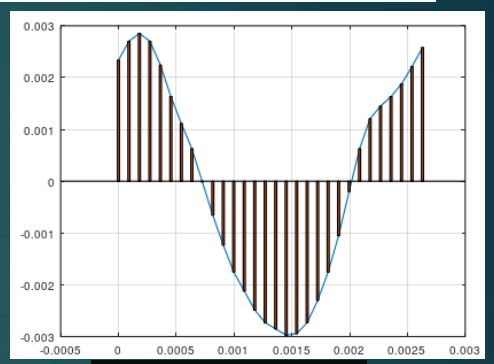
>> ini=6238;



>> ini=10238;

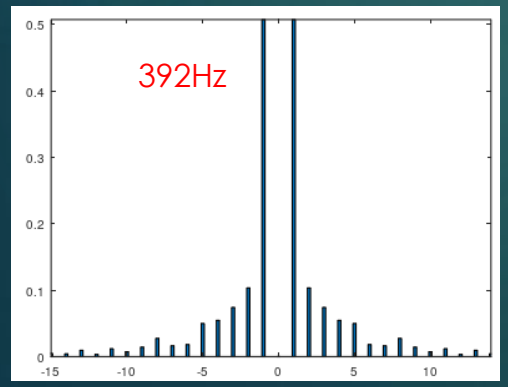


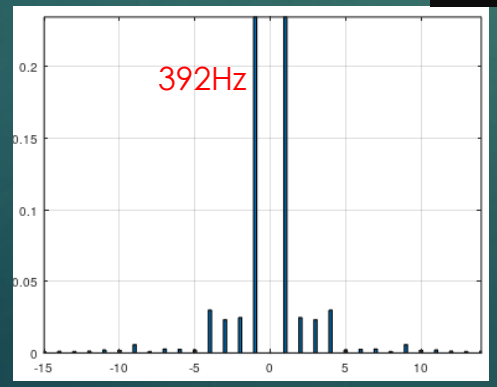
>> ini=20244;

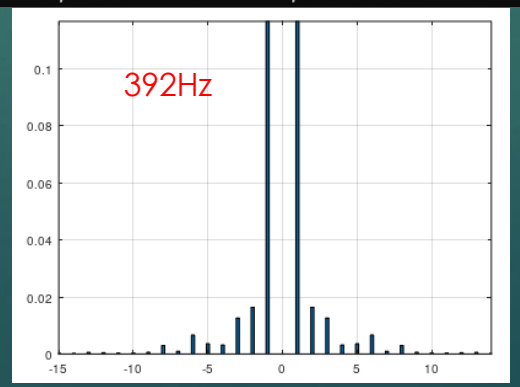


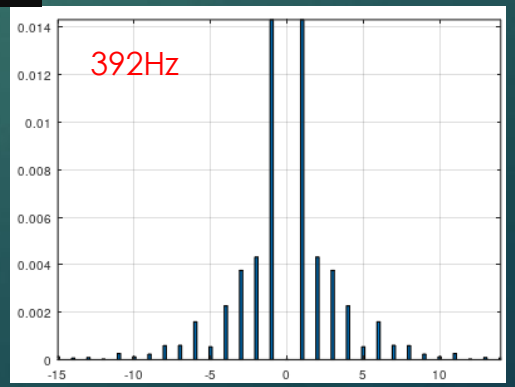
>> ini=28252;

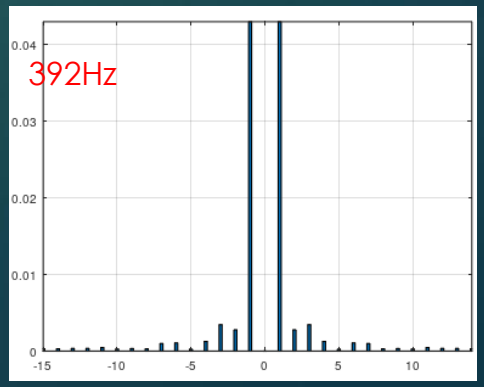
```
>> ini=2267;
>> Y=fft(y(ini:ini+Np)-mean(y(ini:ini+Np)));
>> n=[-(Np-1)/2-1:(Np-1)/2];
>> bar(n,fftshift(abs(Y)),0.2)
```









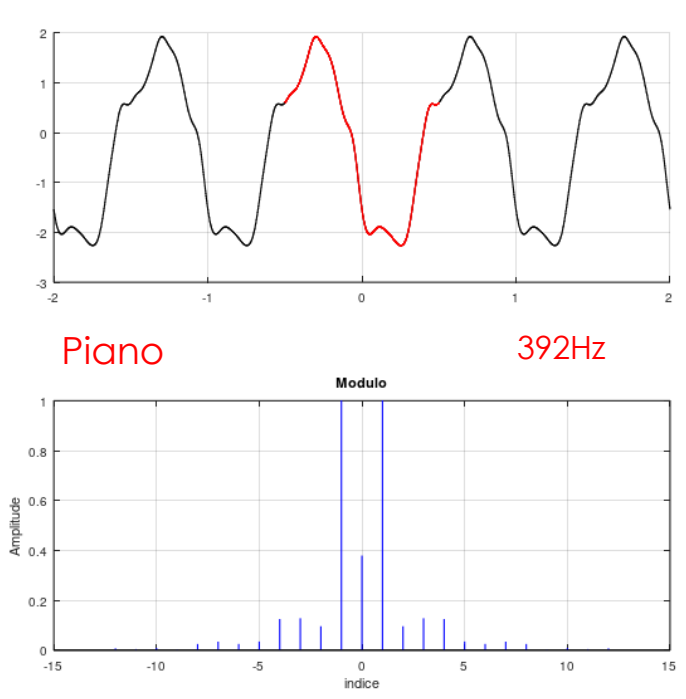


Com o auxílio da FFT, que será vista no Módulo 12, calcula 14 harmônicos da Série de Fourier Média de um sinal periódico y1 com frequência central fc, amostrado com frequência fs

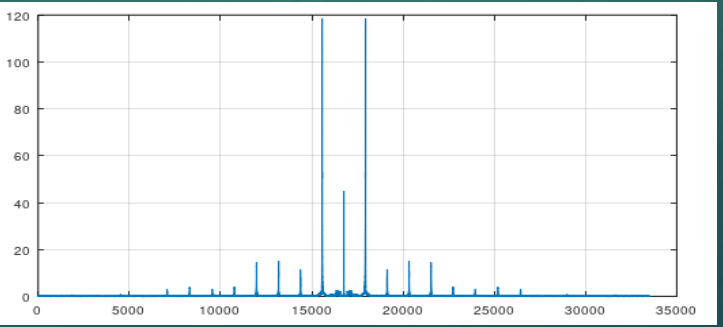
```
function F=SerieFourierMediaSinal(y1,fs,fc)
    Y1=fft(y1);
    mY1=abs(Y1)/max(abs(Y1));
    fY1=atan2d(imag(Y1),real(y1));
    N=length(y1);
    f0=fs/N;
    delta=floor(fc/f0);
    k=1;
    for indx=1:delta:N/2-delta/2
        ini = floor(indx-delta/2);
        if (ini<1) ini = 1; end
        [mag indx2]=max(mY1(ini:floor(indx-1+delta/2)));
        fase=fY1(indx2+ini-1);
        F(1,k)=(mag*exp( i *fase));
        k=k+1;
        if(k==14) return; end
    end
endfunction
```

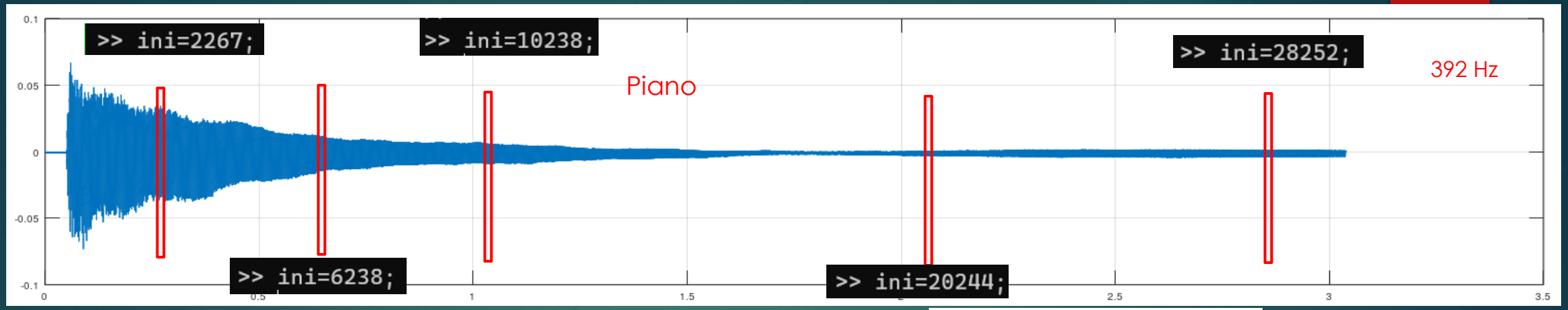
```
function PlotSerieFourierMediaSinal(y1,fs,fc)
    clf;
    F=SerieFourierMediaSinal(y1,fs,fc);
    N=length(F);
    F=[conj(flip(F(2:N))) F];
    T=1;
    w0 = 2* pi  /T;
   subplot(2,1,1)
    t3=linspace(-2*T,2*T,2000);
    y3=SinteseSerieFourier(F,t3,-T/2,T/2);
    hold on; grid on;
    plot(t3,y3,'bk');
    t1=linspace(-T/2,T/2,2000);
    y1=SinteseSerieFourier(F,t1,-T/2,T/2);
    plot(t1,y1,'r');
   subplot(2,1,2)
    n=[-N+1:N-1];
    stem(n,abs(F),'b','marker','none'); % modulo
    hold on; grid on;
    title("Modulo");xlabel("indice");ylabel("Amplitude");
endfunction
```

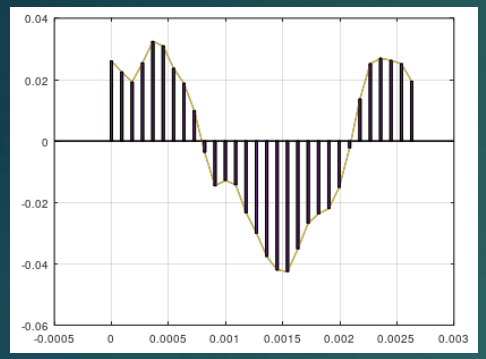
```
>> [y,fs]=audioread("./Dados/piano.wav");
>> PlotSerieFourierMediaSinal(y,fs,392);
```

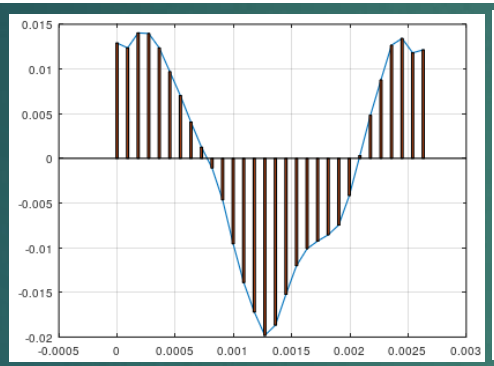


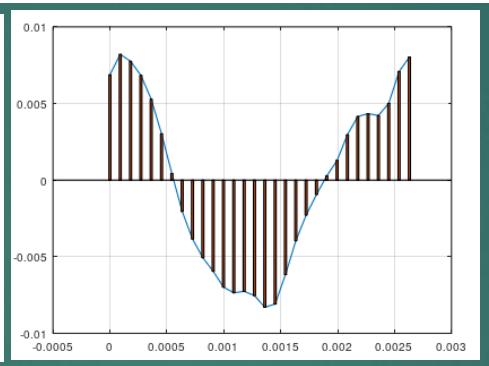

```
>> [y,fs]=audioread("./Dados/piano.wav");
>> plot(abs(fftshift(fft(y))));
```

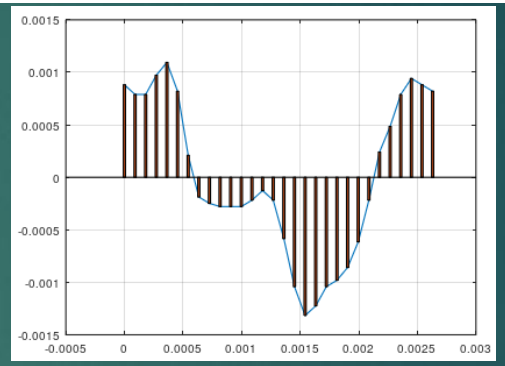


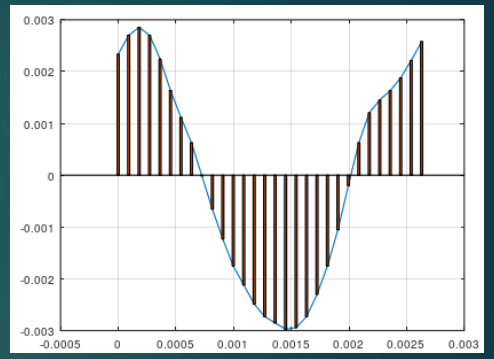




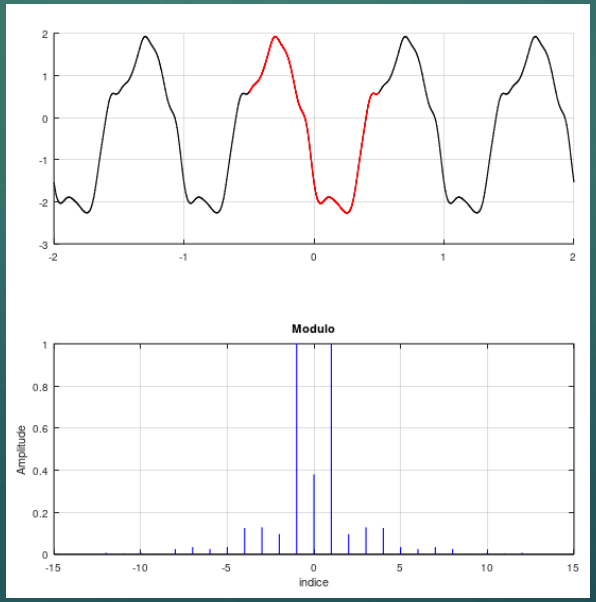




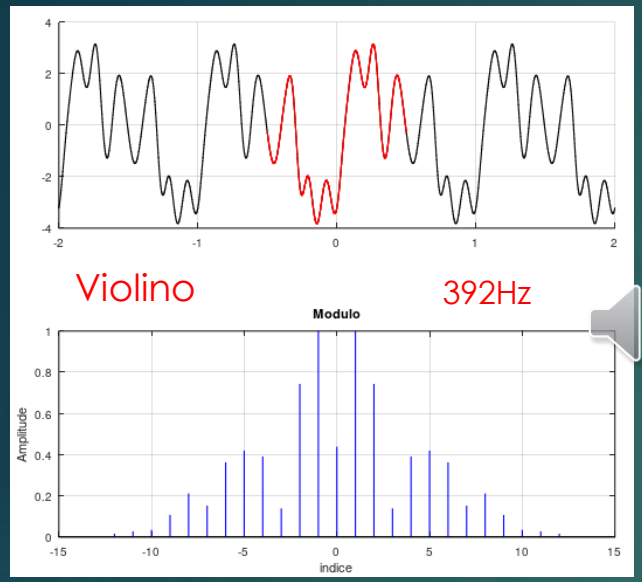




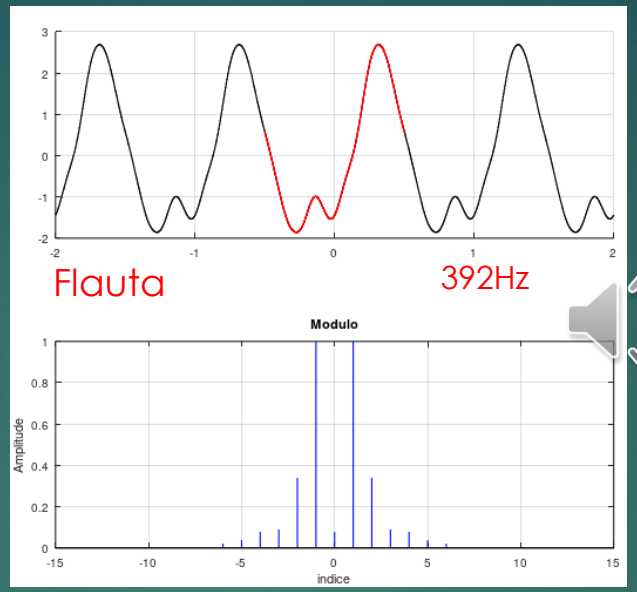
```
>> [y,fs]=audioread("./Dados/piano.wav");
>> PlotSerieFourierMediaSinal(y,fs,392);
```



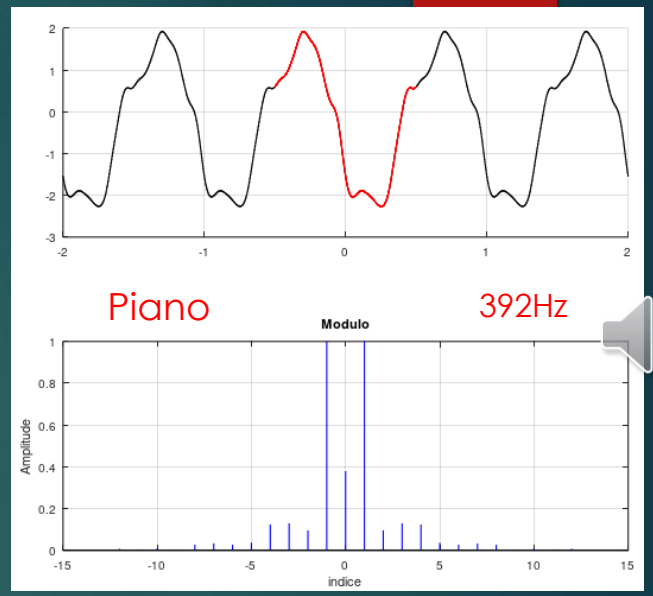
```
>> [y,fs]=audioread("./Dados/violino.wav");
>> PlotSerieFourierMediaSinal(y,fs,392);
```



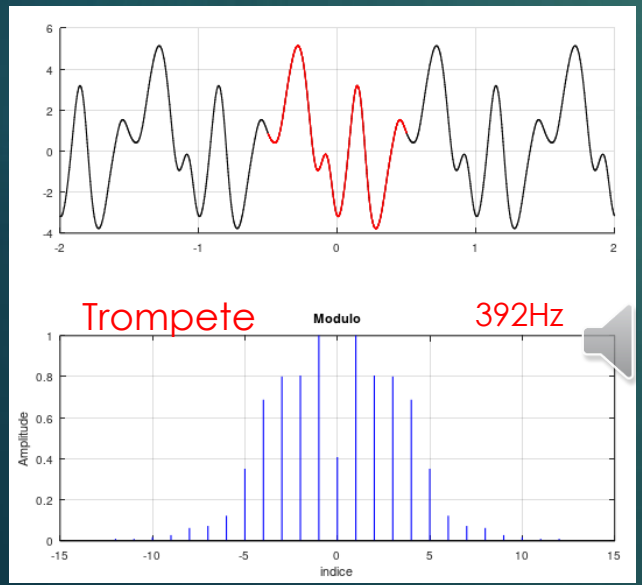

```
>> [y,fs]=audioread("./Dados/flauta.wav");
>> PlotSerieFourierMediaSinal(y,fs,392);
```



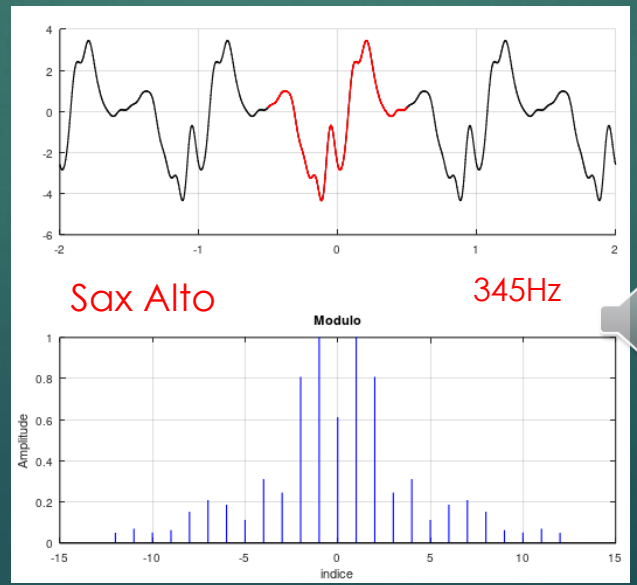

```
>> [y,fs]=audioread("./Dados/piano.wav");
>> PlotSerieFourierMediaSinal(y,fs,392)
```



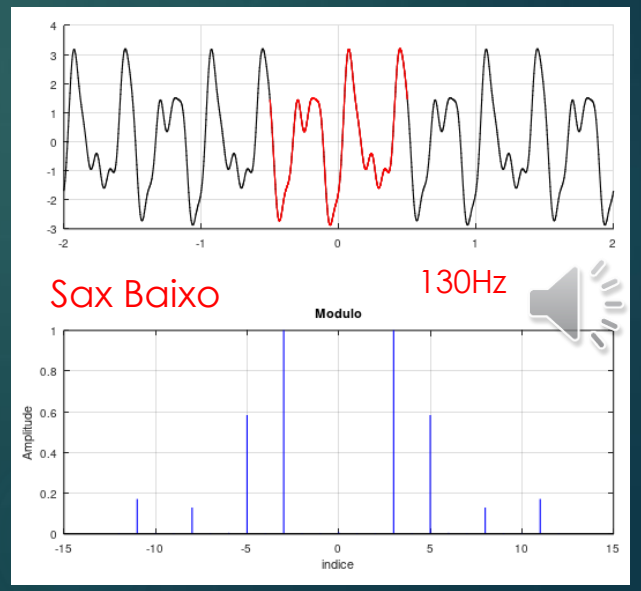

```
>> [y,fs]=audioread("./Dados/trompete.wav");
>> PlotSerieFourierMediaSinal(y,fs,392);
```




```
>> [y,fs]=audioread("./Dados/saxofone_alto.wav");
>> PlotSerieFourierMediaSinal(y,fs,345);
```




```
>> [y,fs]=audioread("./Dados/saxofone_baixo.wav");
>> PlotSerieFourierMediaSinal(y,fs,130);
```



Exemplo 11 - Trem de pulsos retangulares

- CASO GERAL
- Período "T" $\left(w_0 = \frac{2\pi}{T}\right)$; Largura do pulso "a"

Solução analítica para a Série

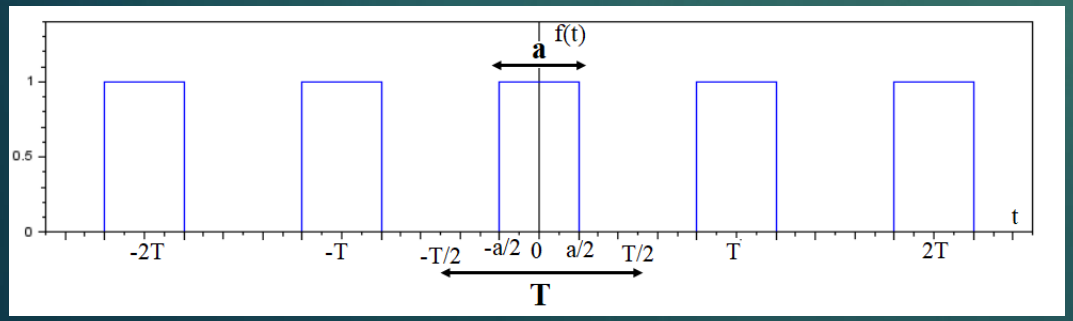
$$f(t) = u(t + a/2) - u(t - a/2);\ \ período\ T$$

$$F[n] = \frac{1}{T}\int_0^T f(t)e^{-jnw_o t}dt = \frac{1}{T}\int_{-\frac{a}{2}}^{\frac{a}{2}}(1)e^{-jnw_o t}dt$$

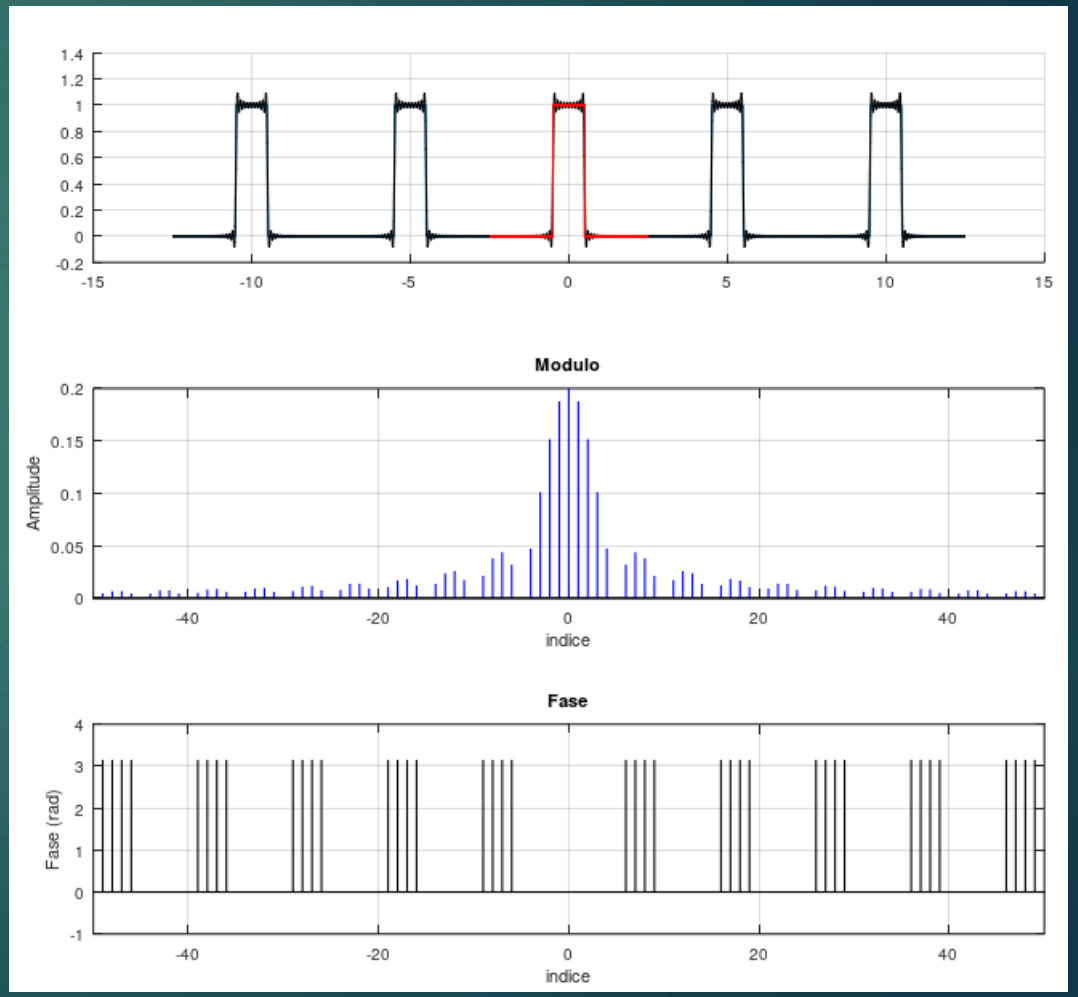
$$F[n] = \frac{1}{T}\left(\frac{1}{-jnw_o}\right)e^{-jnw_o t}\Bigg|_{-\frac{a}{2}}^{+\frac{a}{2}} = \left(\frac{w_o}{2\pi}\right)\left(\frac{1}{-jnw_o}\right)\left[e^{-\frac{-jnw_o a}{2}} - e^{-\frac{jnw_o a}{2}}\right]$$

$$F[n] = \frac{a\,w_o}{2\pi}\frac{\sin\left(\frac{nw_o a}{2}\right)}{\frac{nw_o a}{2}} = \frac{a}{T}sinc\left(\frac{nw_o a}{2}\right)$$

```
>> a=1;
>> f1 = @(t) f_u(t+a/2)-f_u(t-a/2);
>> T=5;
>> t=linspace(-2.5*T,2.5*T,1001);
>> plot(t,MakePeriodic(f1,t,-T/2,T/2),'b')
```




```
>> a=1;
>> T=5;
>> f1 = @(t) f_u(t+a/2)-f_u(t-a/2);
>> PlotSerieFourier(f1,-T/2,T/2,50)
```

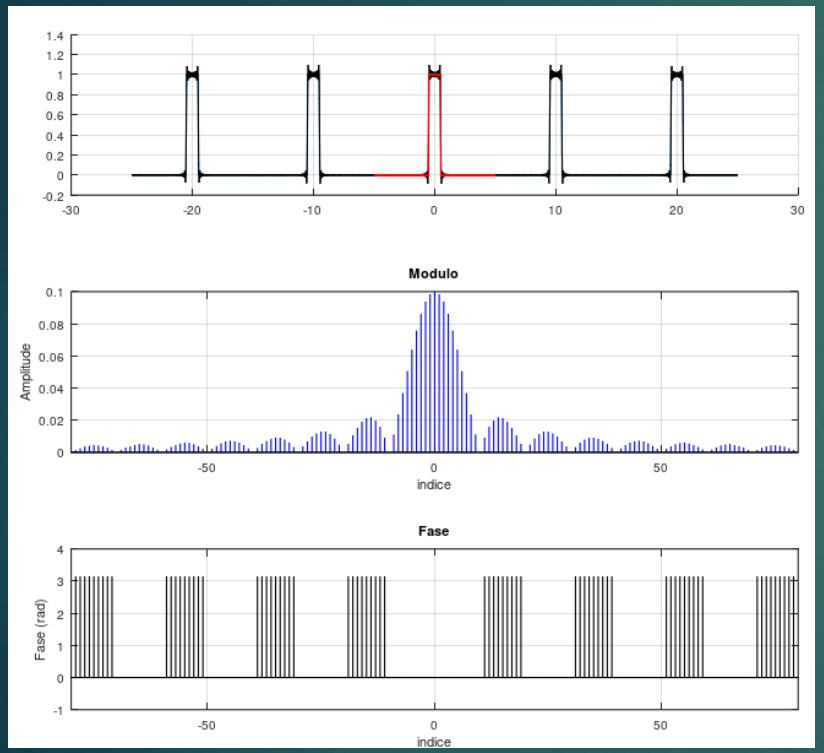


```
--> deff('y=f(t)','y=f_u(t+a/2)-f_u(t-a/2)');
```

$$F[n] = \frac{a}{T} sinc\left(\frac{n\pi w_o a}{2}\right)$$

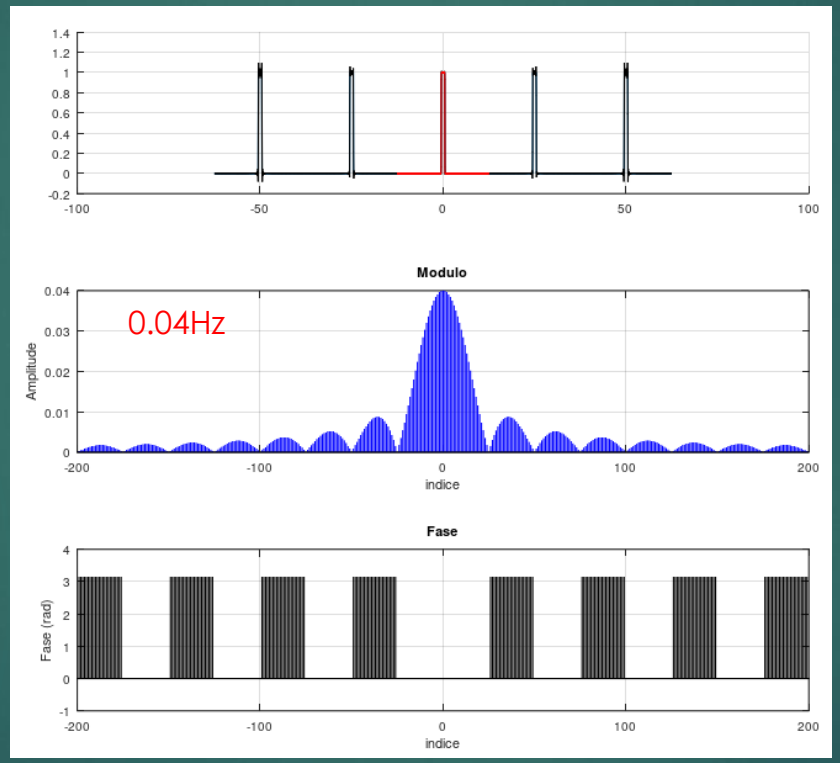
a=1 T=10

```
>> a=1;
>> T=10;
>> PlotSerieFourier(f1,-T/2,T/2,80)
```



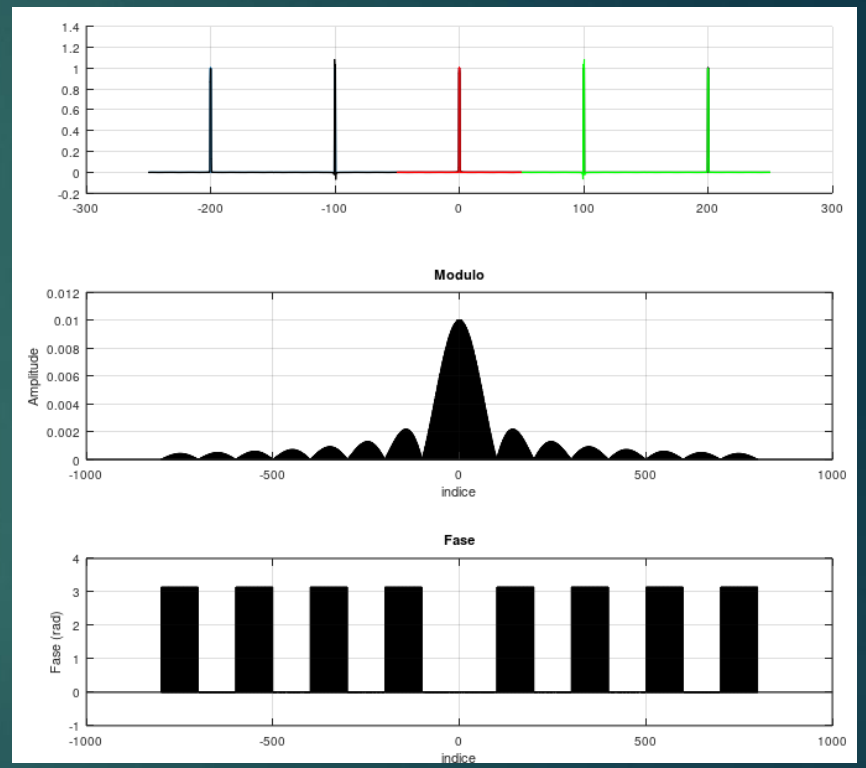
a=1 T=25

```
>> a=1;
>> T=25;
>> PlotSerieFourier(f1,-T/2,T/2,200)
```



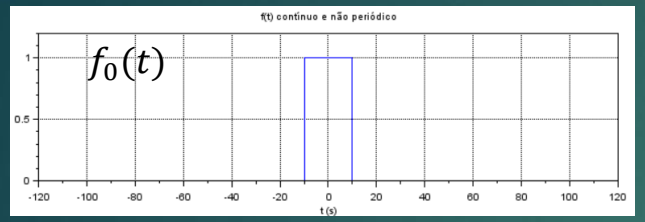
a=1 T=100

```
>> a=1;
>> T=100;
>> PlotSerieFourier(f1,-T/2,T/2,800)
```

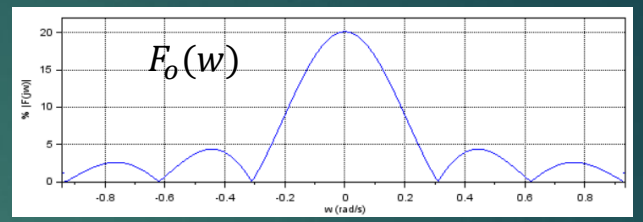


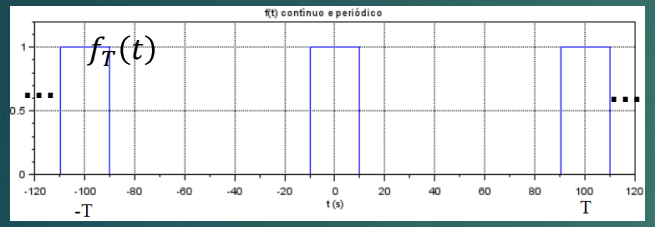
Notar que quando o período T aumenta , o espaçamento entre os harmônicos ($w_o = \frac{2\pi}{T}$) diminui, e também diminuem os valores absolutos dos coeficientes $F[n] \sim a/T$.
Quando T aumenta, notamos que a envoltória dos coeficientes converge para um $sinc(x)$ contínuo

Chegamos à Série de Fourier $F(n)$, tornando periódico o sinal $f_0(t\ )$ que era aperiódico, e calculando no limite a sua Transformada de Fourier $F_o(w)$

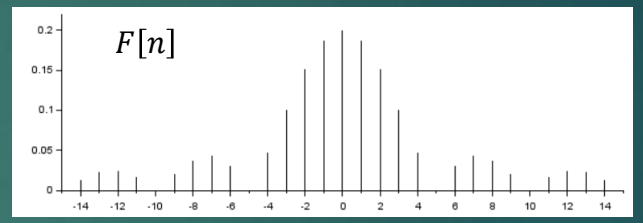


$$\xRightarrow{Transf.\,de\,Fourier}$$



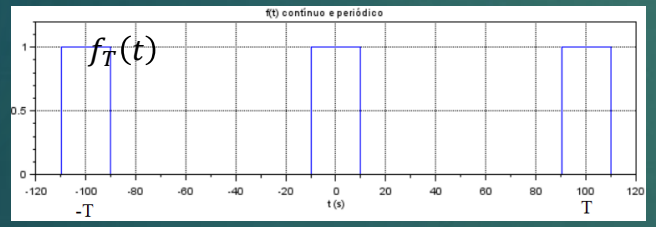


$$\xRightarrow{Transf.\,de\,Fourier}$$

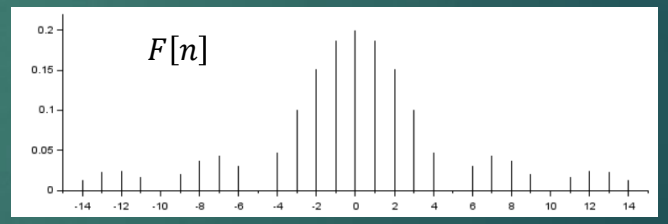


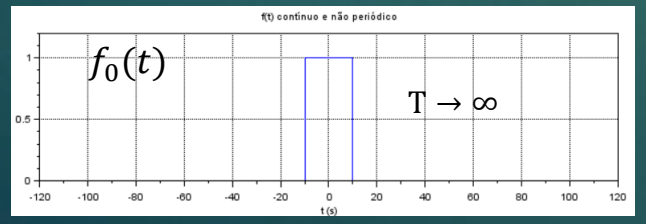
No limite quando o sinal se torna periódico teremos a série de Fourier

Podemos voltar à Transformada de Fourier $F_o(w)$, tornando aperiódico o sinal $f_T(t)$ que era periódico, e calculando no limite a sua Série de Fourier $F(n)$

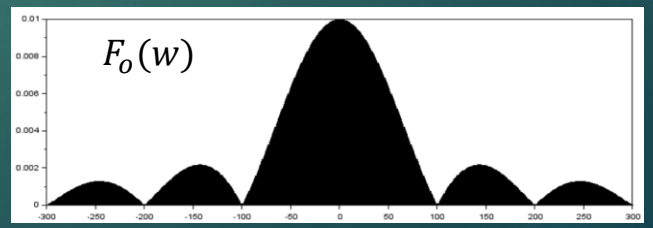


$$\xRightarrow{Série\,de.\,Fourier}$$





$$\xRightarrow{Série\,de.\,Fourier}$$



No limite quanto $T \to \infty$ Teremos a Transformada de Fourier

Obtendo a Transformada de Fourier através do limite da Série de Fourier $F(n)$ quando $T \rightarrow \infty$

Como o período do trem de pulsos é T, então a frequência fundamental $w_o = \frac{2\pi}{T}$

O espaçamento entre os harmônicos Δw é igual à frequência fundamental w_0

Notar que quando $T \rightarrow \infty$, a função deixa de ser periódica (somente se repete no infinito!)´. A função será então um pulso isolado, que não se repete.

Notar que quanto $T \rightarrow \infty$, $\Delta w \rightarrow 0$, $w_o \rightarrow 0$ e $F(n) \rightarrow 0$!!!

Como o espaçamento $\Delta w \rightarrow 0$, a série discreta de coeficientes F(n) tende a uma função contínua.

Como os coeficientes $|F(n)| \rightarrow 0$ quando $T \rightarrow \infty$

$$F[n] = \frac{1}{T}\int_0^T f(t)e^{-jnw_o t}dt \rightarrow 0 \text{ !!!}$$

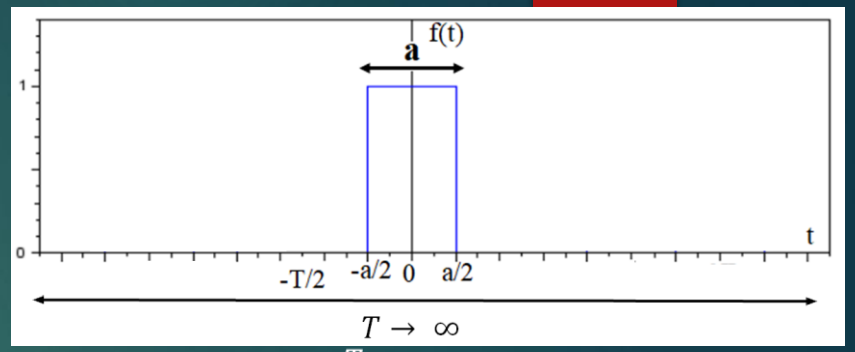
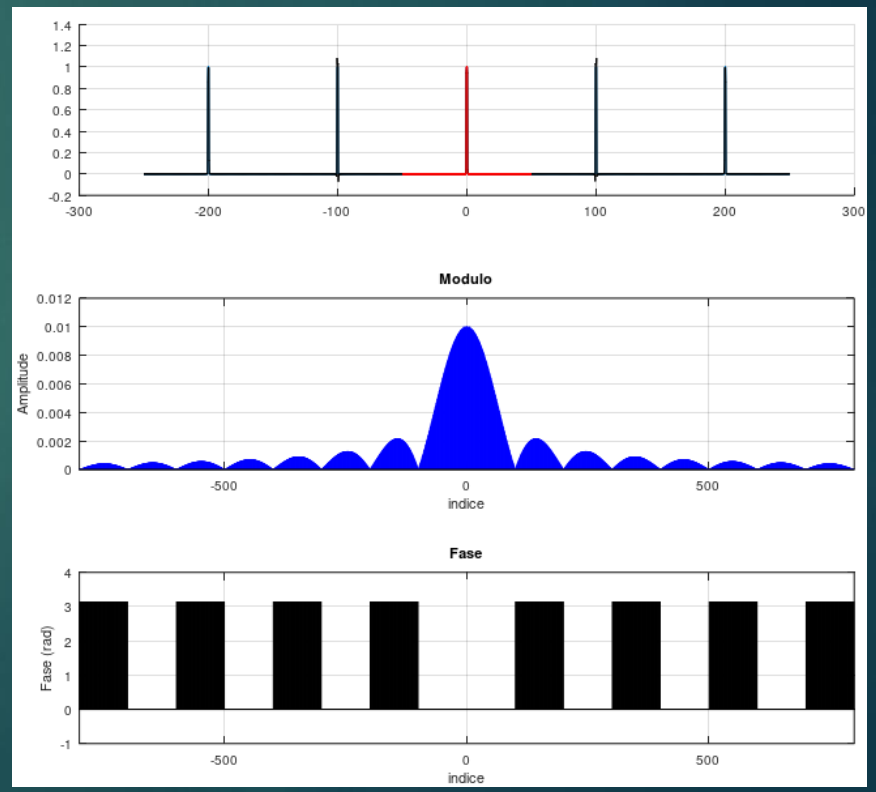
No entanto, o produto $F(n)T = F(jnw_o)$ poderá convergir (zero vezes infinito!):

$$F(jnw_o) = F[n]T = F[n]\frac{2\pi}{w_o} = F[n]\frac{2\pi}{\Delta w} \rightarrow F[n] = F(jnw_o)\frac{\Delta w}{2\pi}$$

$$f(t) = \sum_{n=-\infty}^{\infty} F[n]e^{jnw_o t} = \sum_{n=-\infty}^{\infty}\left(F(jnw_o)\frac{\Delta w}{2\pi}\right)e^{jnw_o t} = \lim_{T\rightarrow\infty}\left[\frac{1}{2\pi}\sum_{n=-\infty}^{\infty} F(jnw_o)e^{jnw_o t}\,\Delta w\right]$$

$$T \rightarrow \infty$$
$$\Delta w \rightarrow dw$$
$$nw_o \rightarrow w$$
$$\sum \rightarrow \int$$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(jw)e^{j\omega t}dw$$



$$F(n) = \frac{1}{T}\int_0^T f(t)e^{-jnw_o t}dt \rightarrow 0 \text{ !!!}$$